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Lintula et al.

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(54) **INTRAMEDULLARY NAIL ALIGNMENT GUIDES, FIXATION GUIDES, DEVICES, SYSTEMS, AND METHODS OF USE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 79 days.

(21) Appl. No.: **18/329,829**

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(65) **Prior Publication Data**

US 2023/0310011 A1 Oct. 5, 2023

Related U.S. Application Data

(60) Continuation of application No. 17/248,165, filed on Jan. 12, 2021, now Pat. No. 11,666,345, which is a (Continued)

(51) **Int. Cl.**
A61B 17/72 (2006.01)
A61B 17/17 (2006.01)

(52) **U.S. Cl.**
CPC **A61B 17/291** (2013.01); **A61B 17/1717** (2013.01); **A61B 17/1725** (2013.01); **A61B 17/1775** (2016.11); **A61B 17/72** (2013.01)

(58) **Field of Classification Search**
CPC . A61B 17/72; A61B 17/7208; A61B 17/7216; A61B 17/7225; A61B 17/7233;
(Continued)

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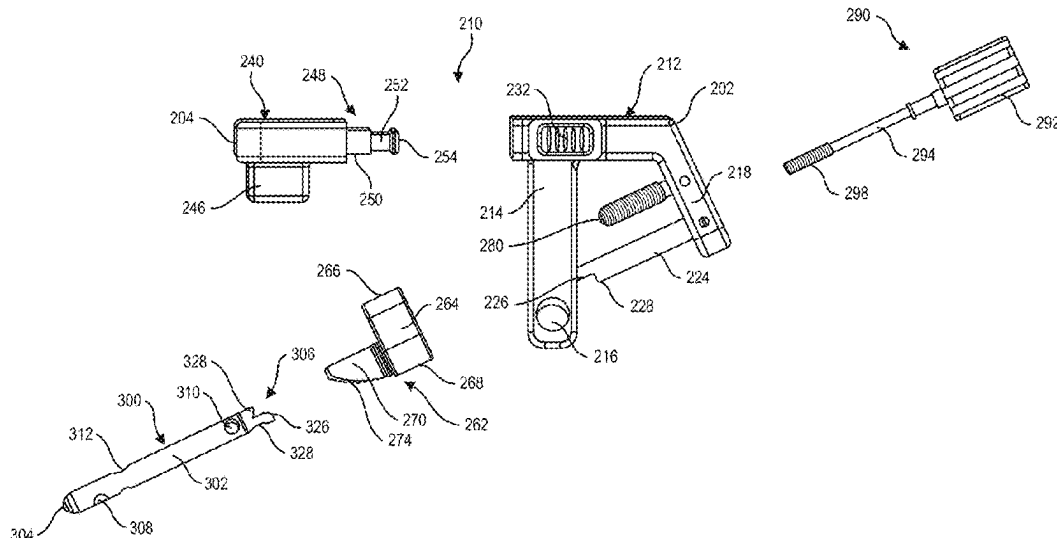
Primary Examiner — Eric S Gibson

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(57) **ABSTRACT**

Alignment guide systems, fixation guide devices, and methods for positioning and inserting an intramedullary nail are disclosed. The alignment guide system includes a targeting guide with a first end and a second end, an alignment wire rotatably engaging a second end of the targeting guide, and a guide sleeve insert engaging an opening in the first end of the targeting guide. The fixation guide device including a frame including a first end and a second end, a compression device slidably coupled to the first end of the frame, and the intramedullary nail secured to a nail attachment apparatus of the frame. A method of inserting an intramedullary nail into two bones for fixation of the two bones is also disclosed.

11 Claims, 30 Drawing Sheets



Related U.S. Application Data

division of application No. 15/907,850, filed on Feb. 28, 2018, now Pat. No. 10,888,338, which is a continuation of application No. PCT/US2018/020046, filed on Feb. 27, 2018.

- (60) Provisional application No. 62/464,175, filed on Feb. 27, 2017.

(58) **Field of Classification Search**

CPC A61B 17/7241; A61B 17/725; A61B 17/7283; A61B 17/7291
See application file for complete search history.

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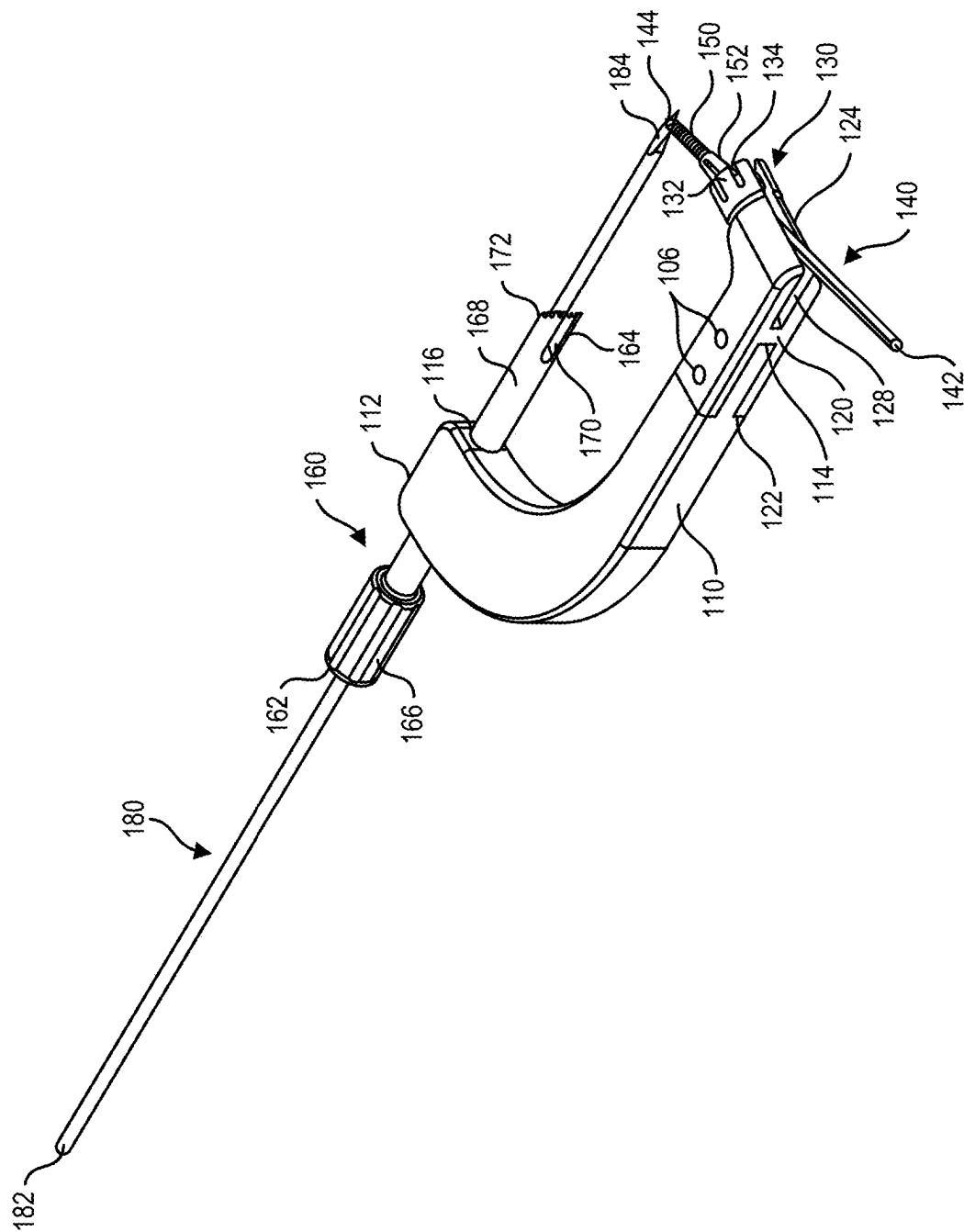


FIG. 1

100

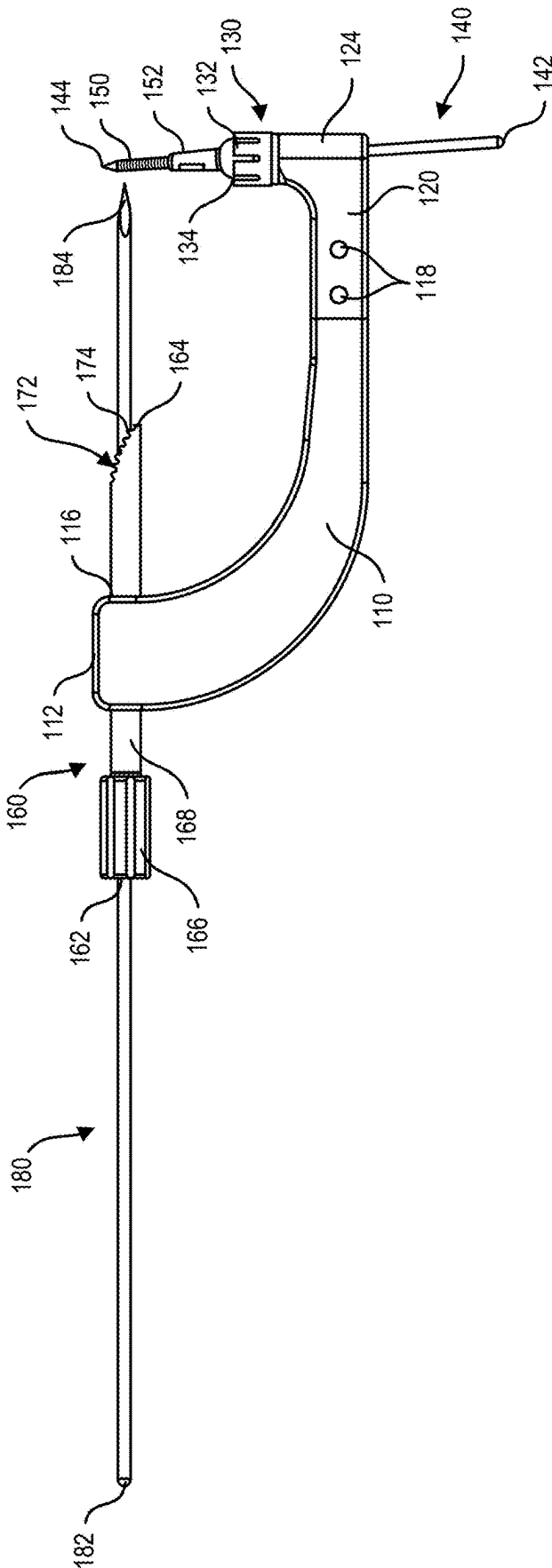


FIG. 2

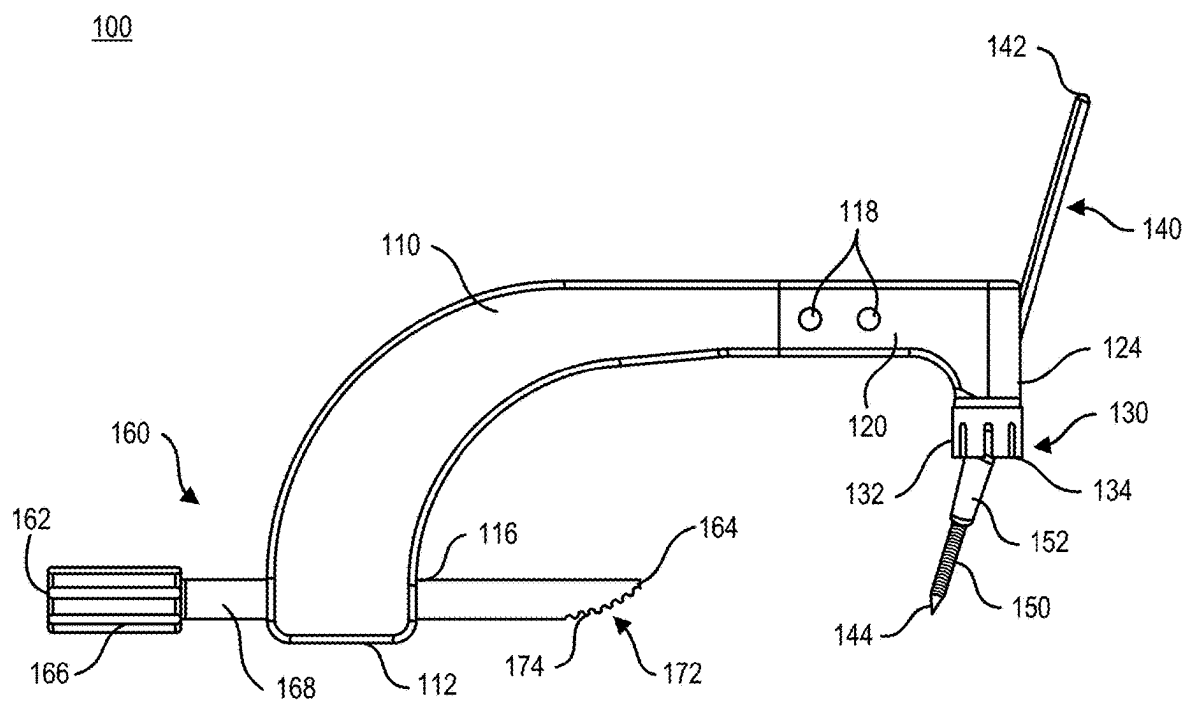


FIG. 3

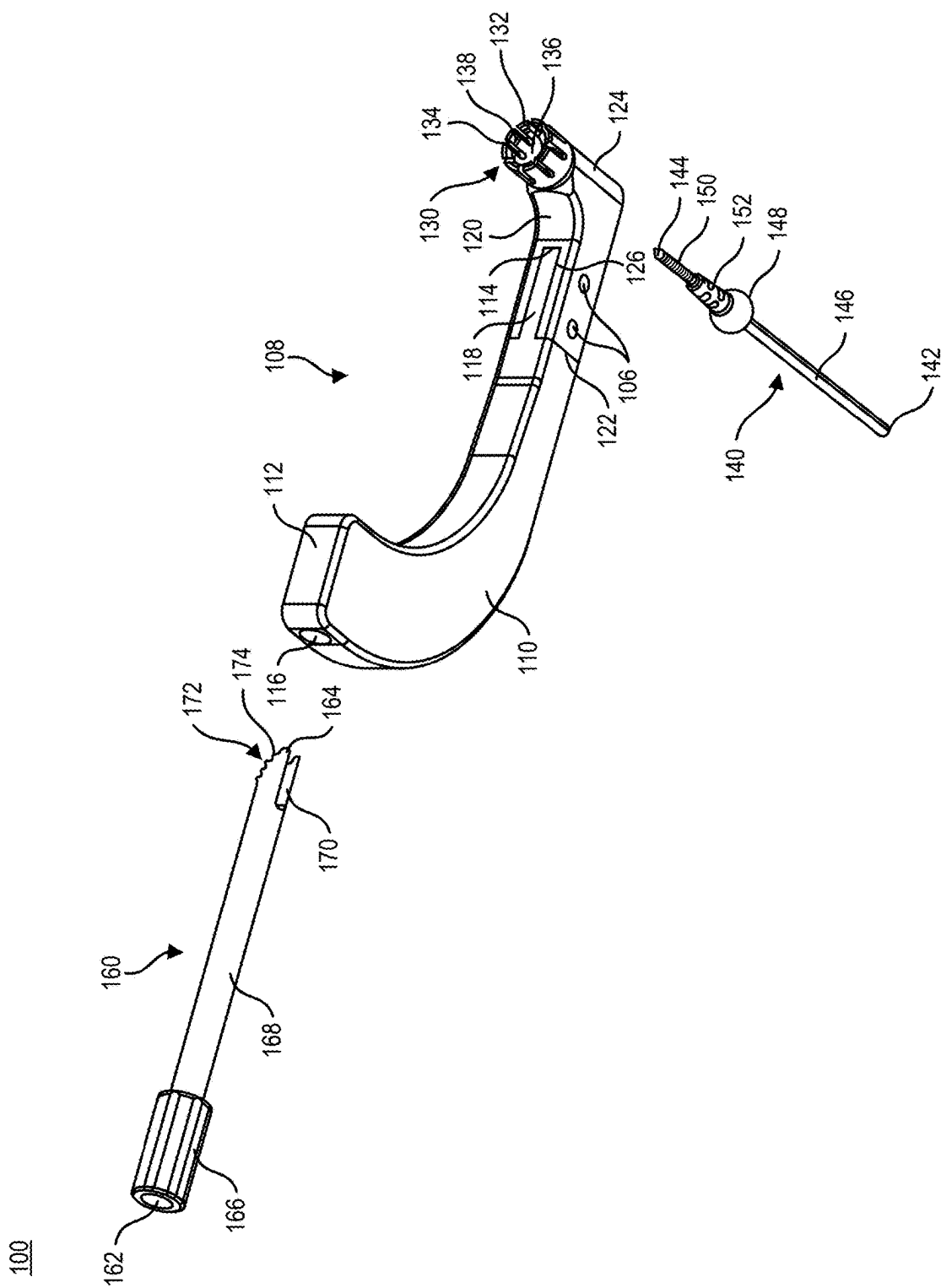


FIG. 5

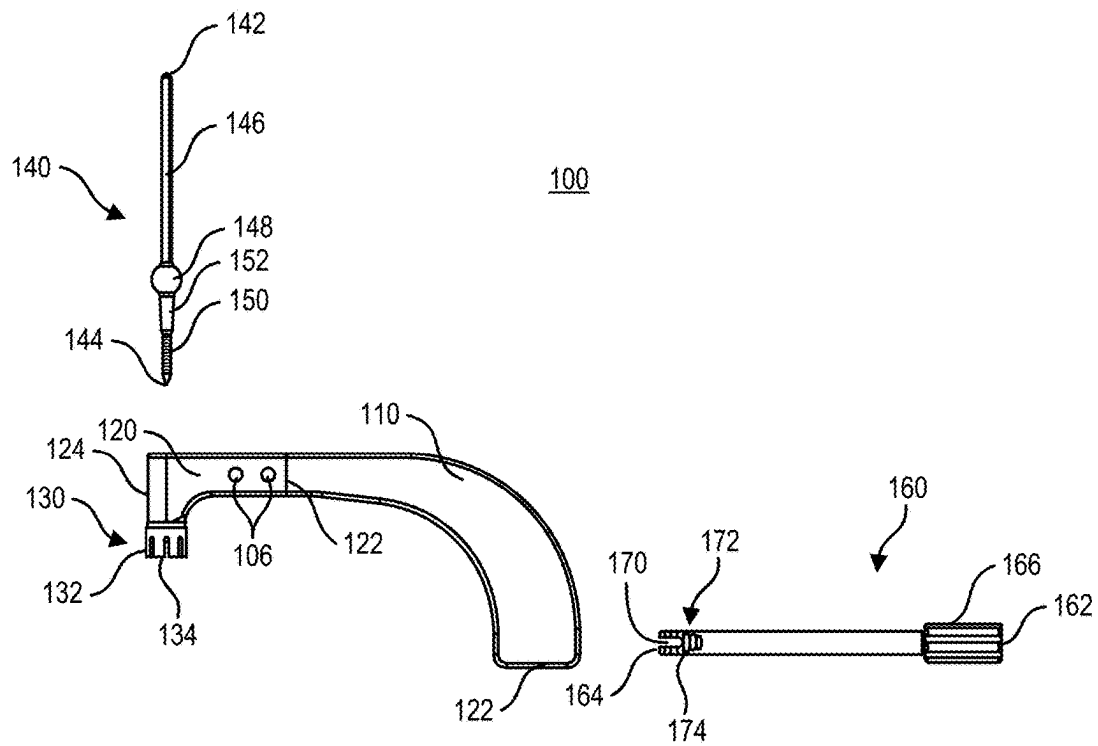


FIG. 6

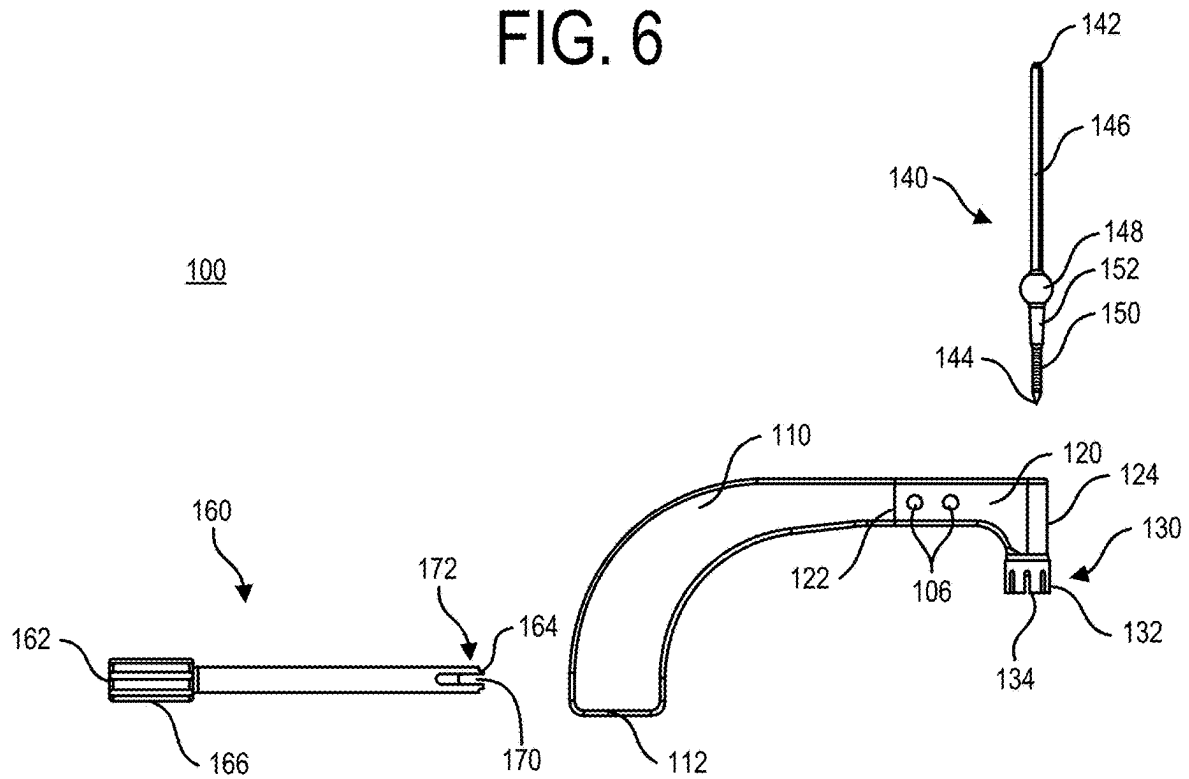


FIG. 7

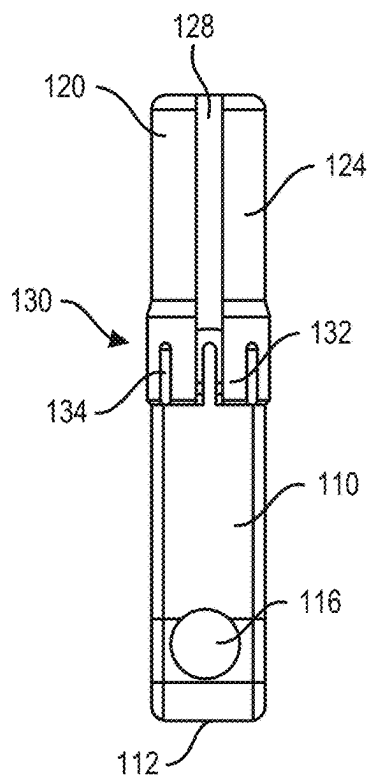


FIG. 8

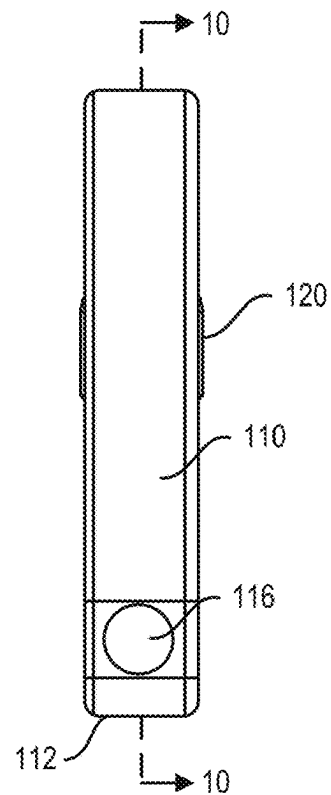


FIG. 9

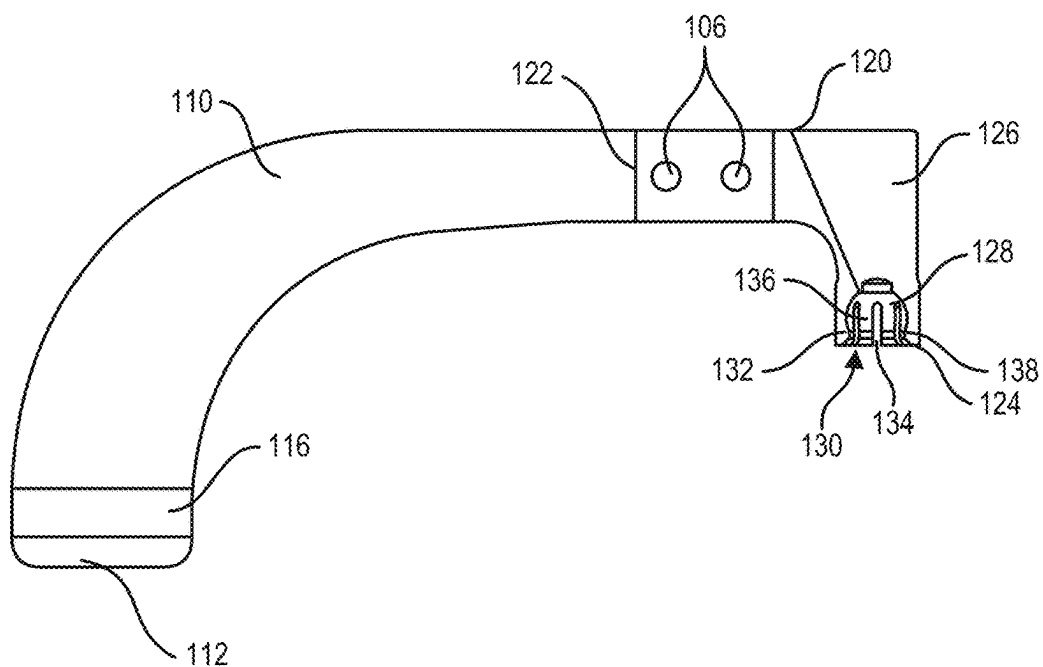


FIG. 10

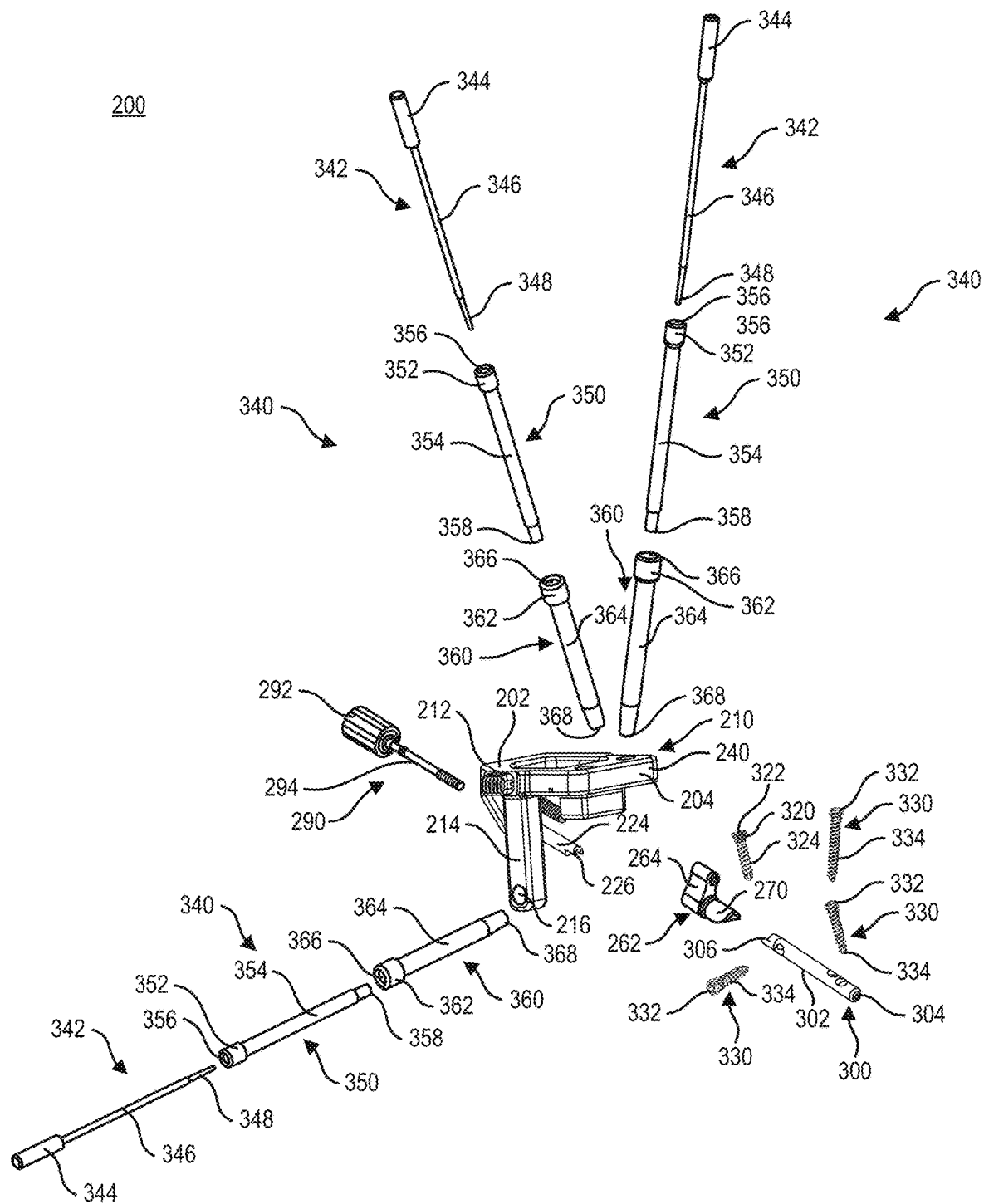
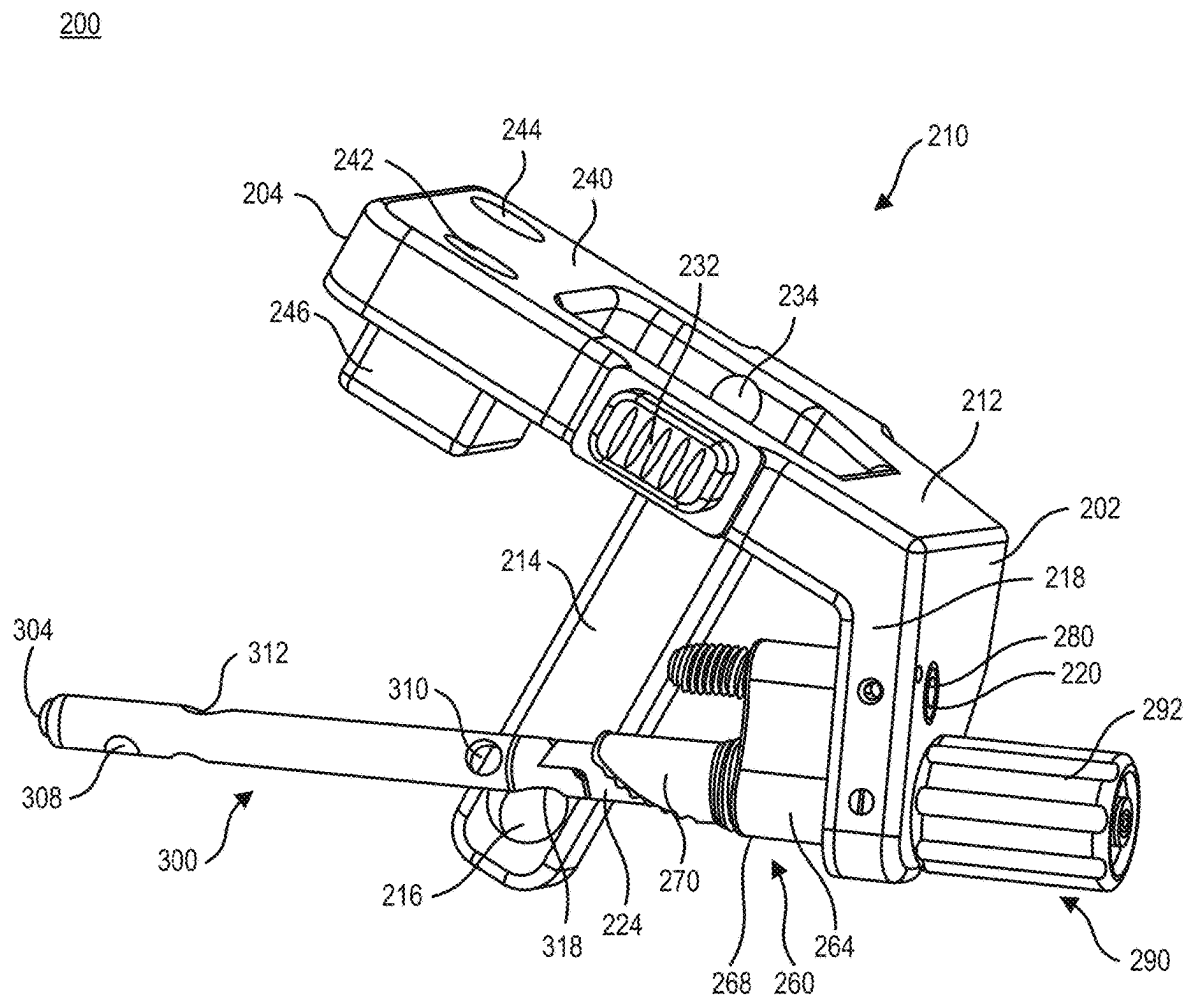


FIG. 11



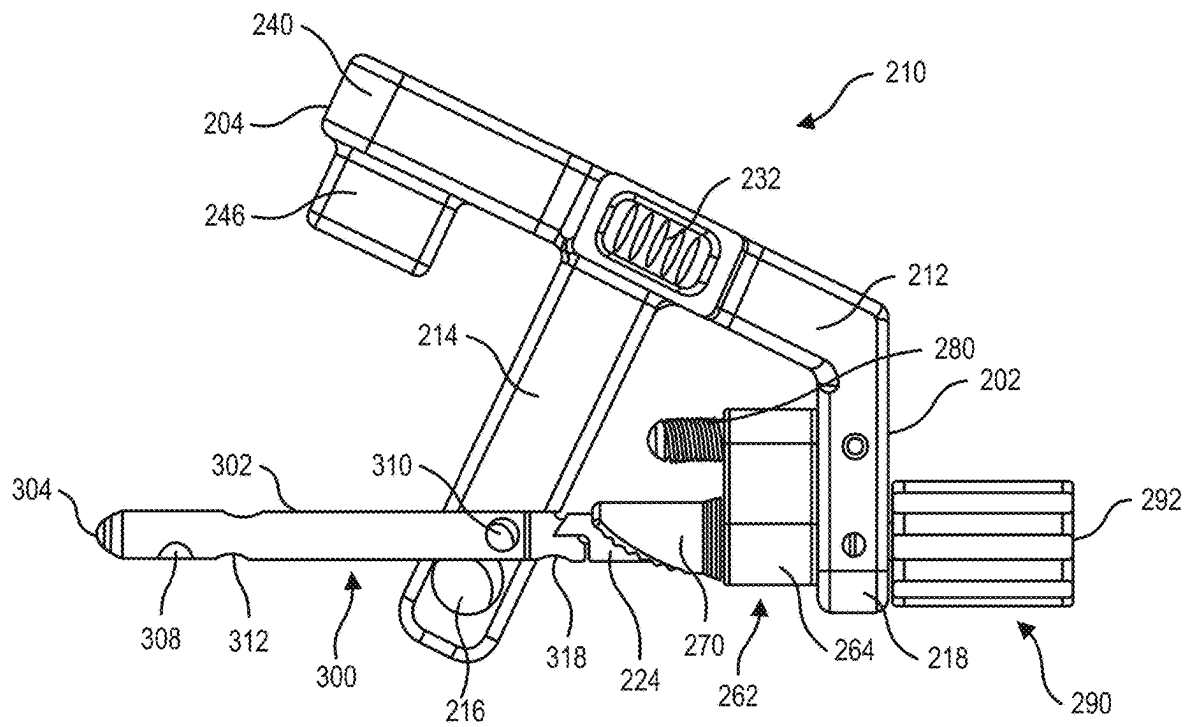


FIG. 13

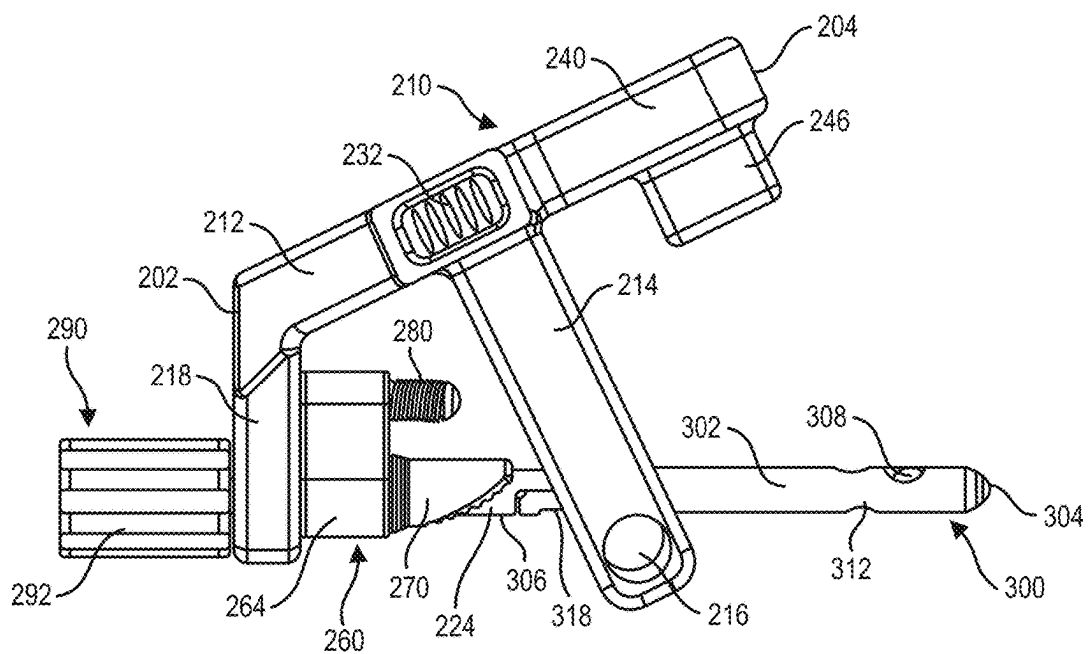


FIG. 14

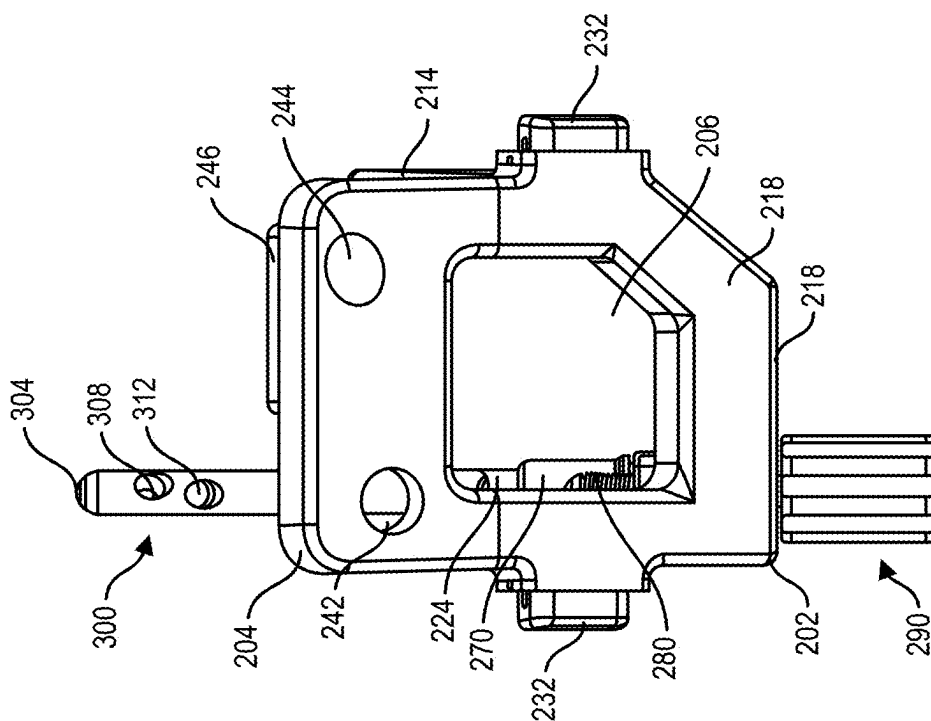


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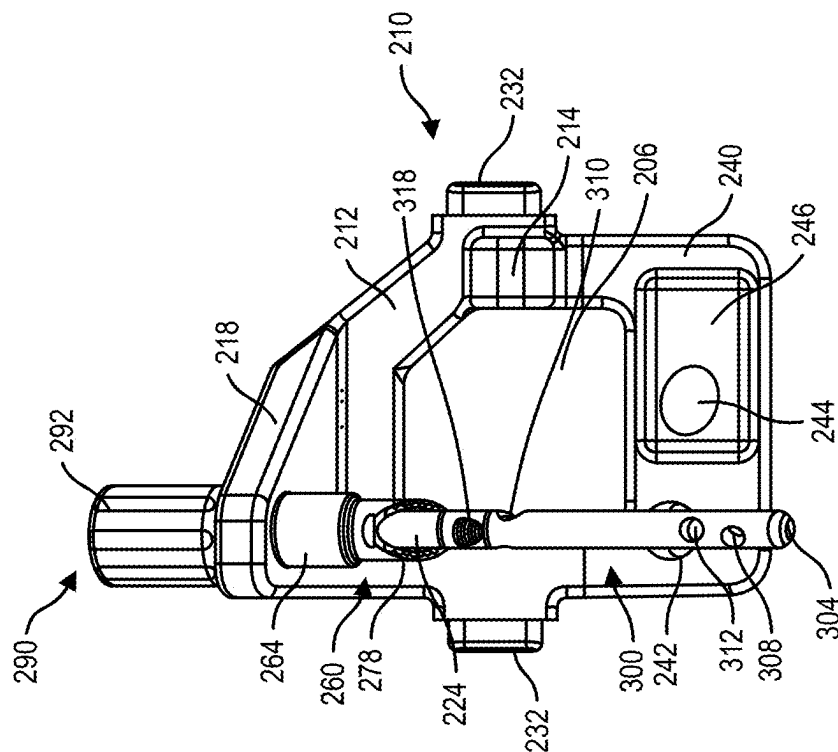


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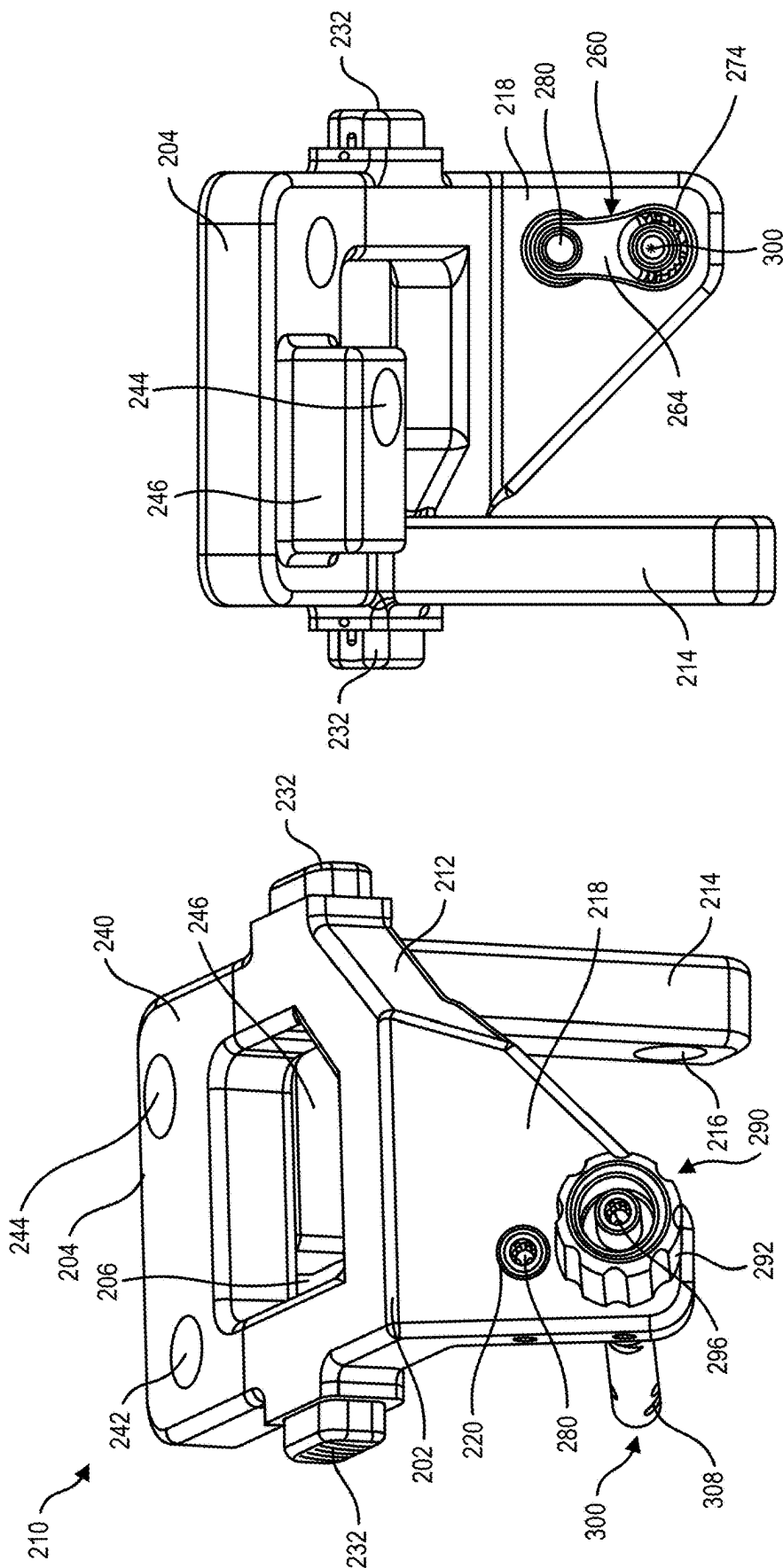


FIG. 18

FIG. 17

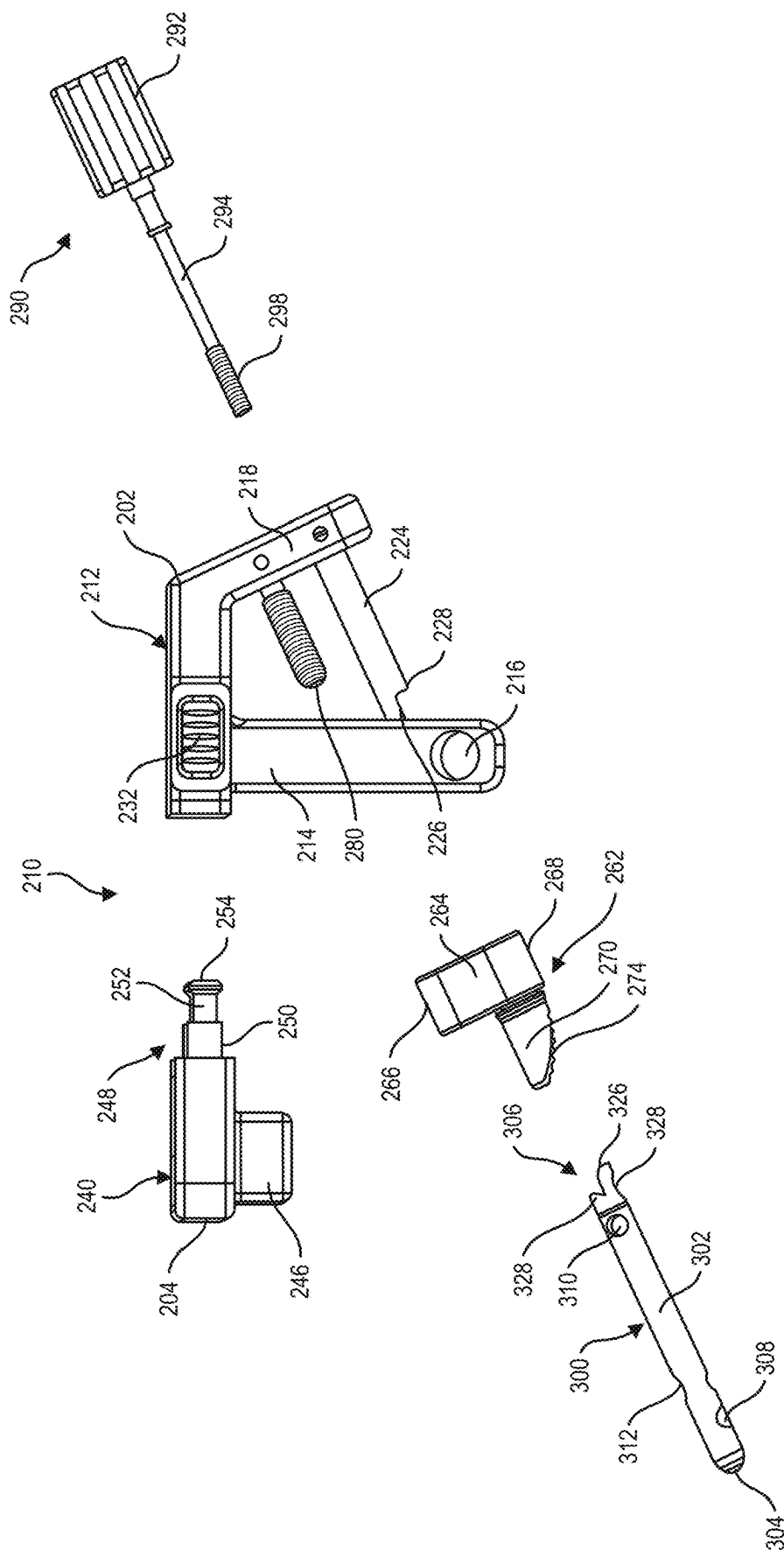


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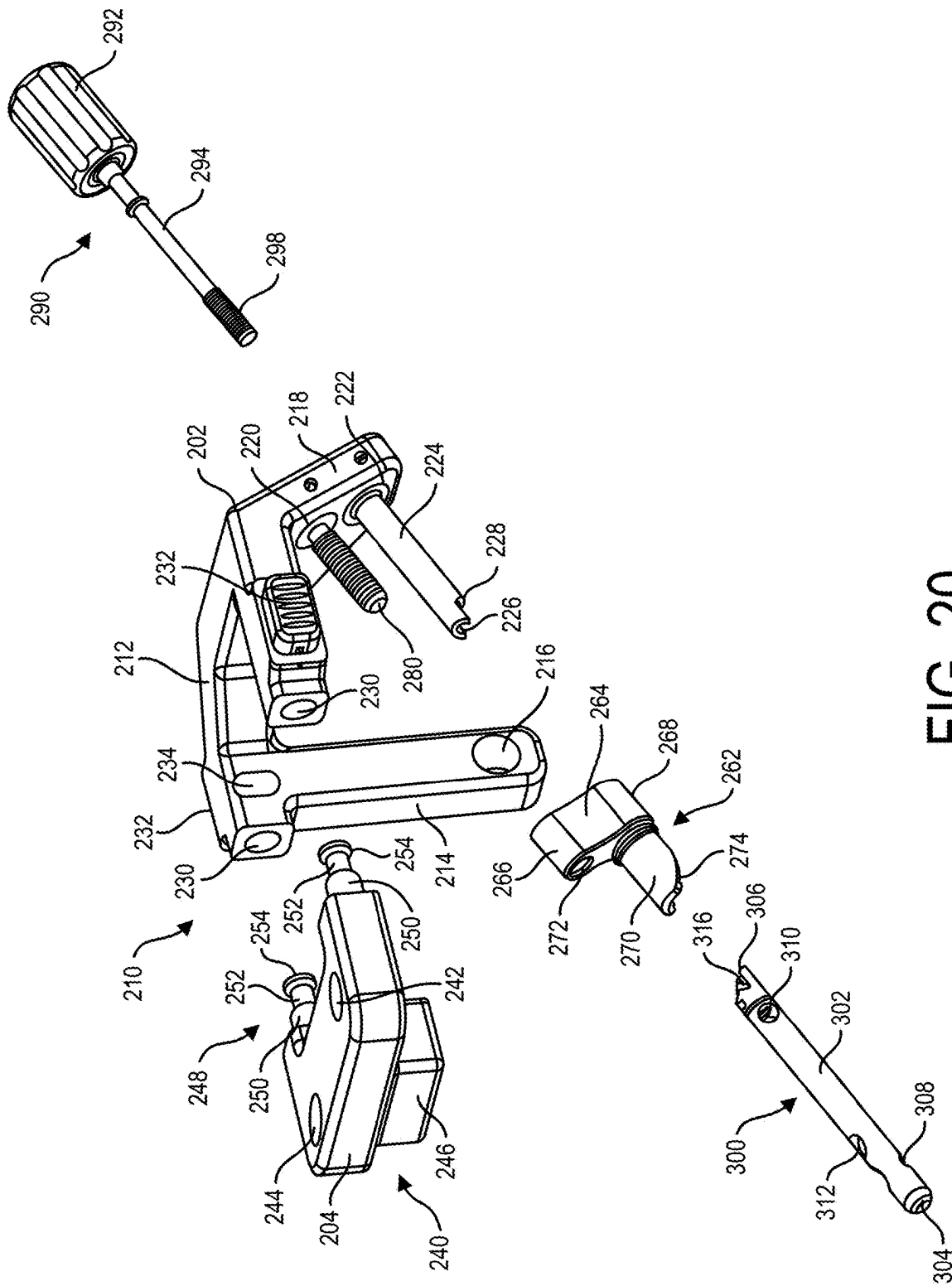


FIG. 20

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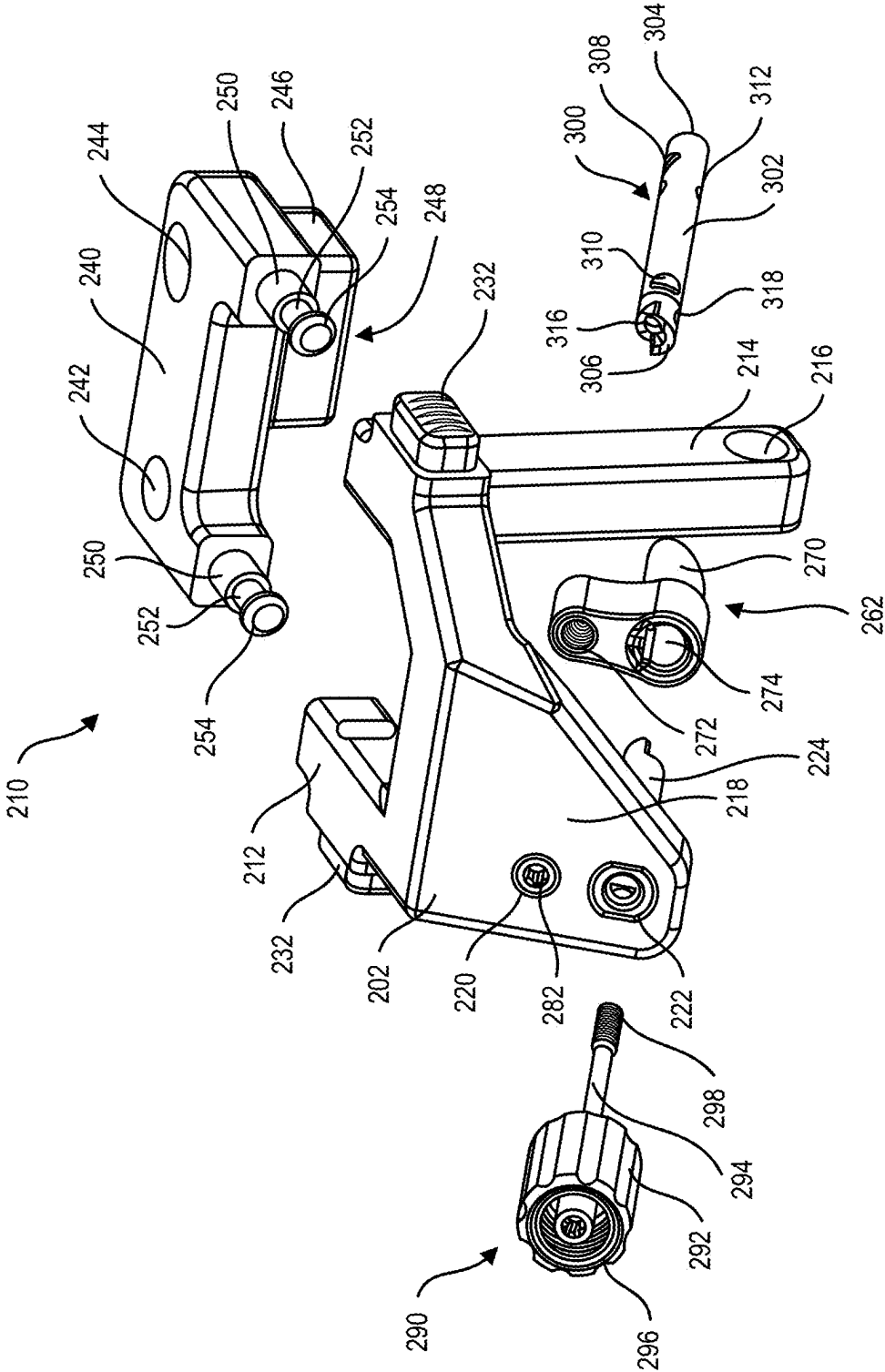


FIG. 21

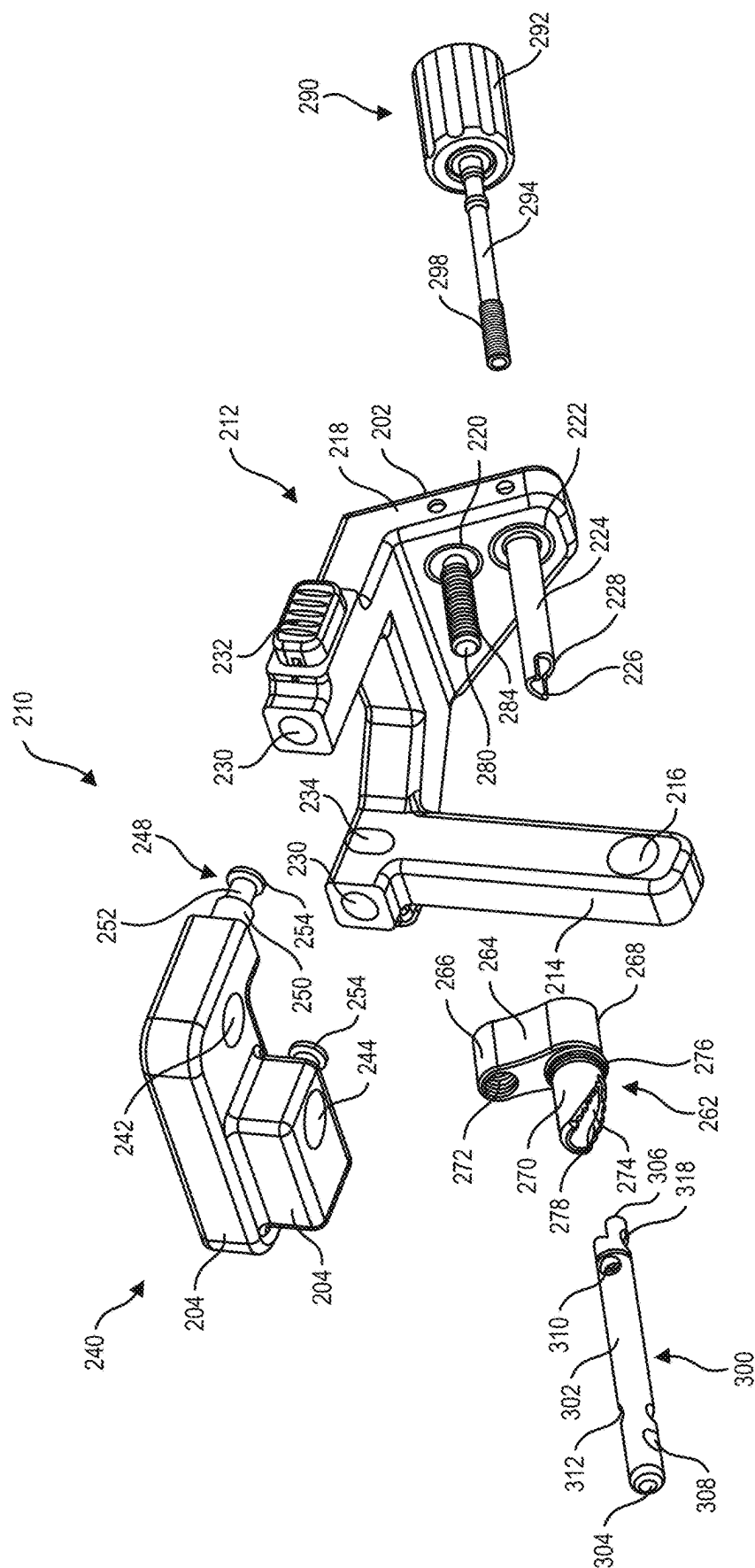


FIG. 22

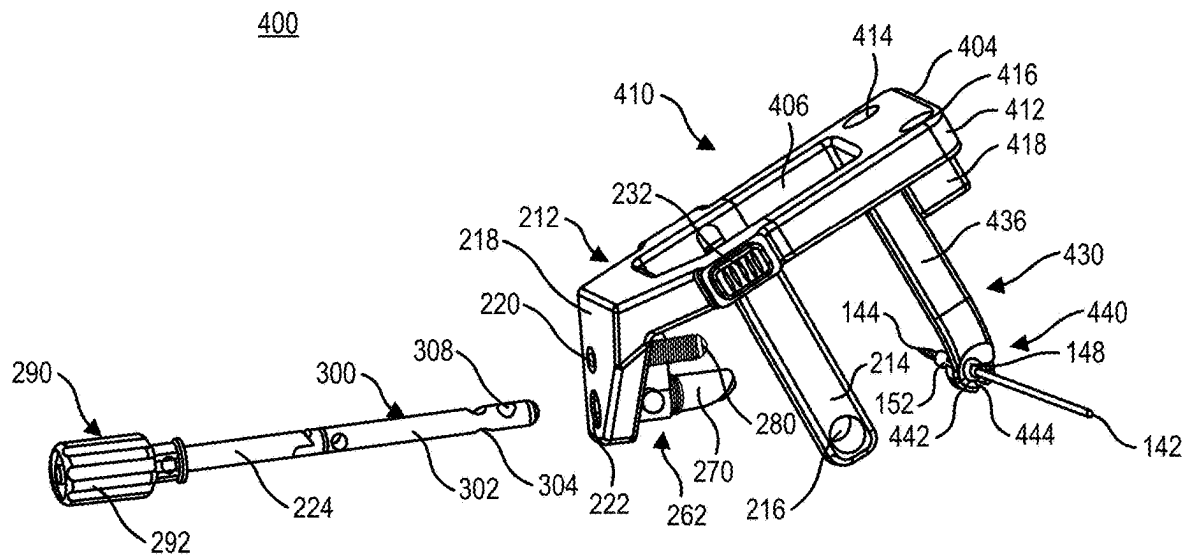


FIG. 23

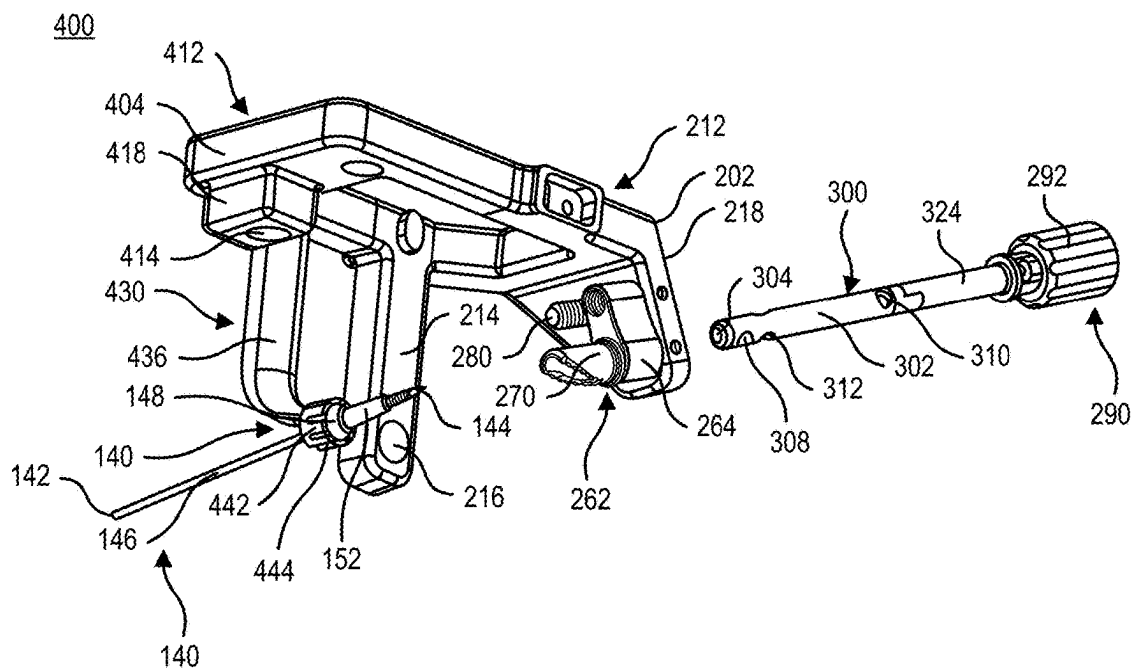


FIG. 24

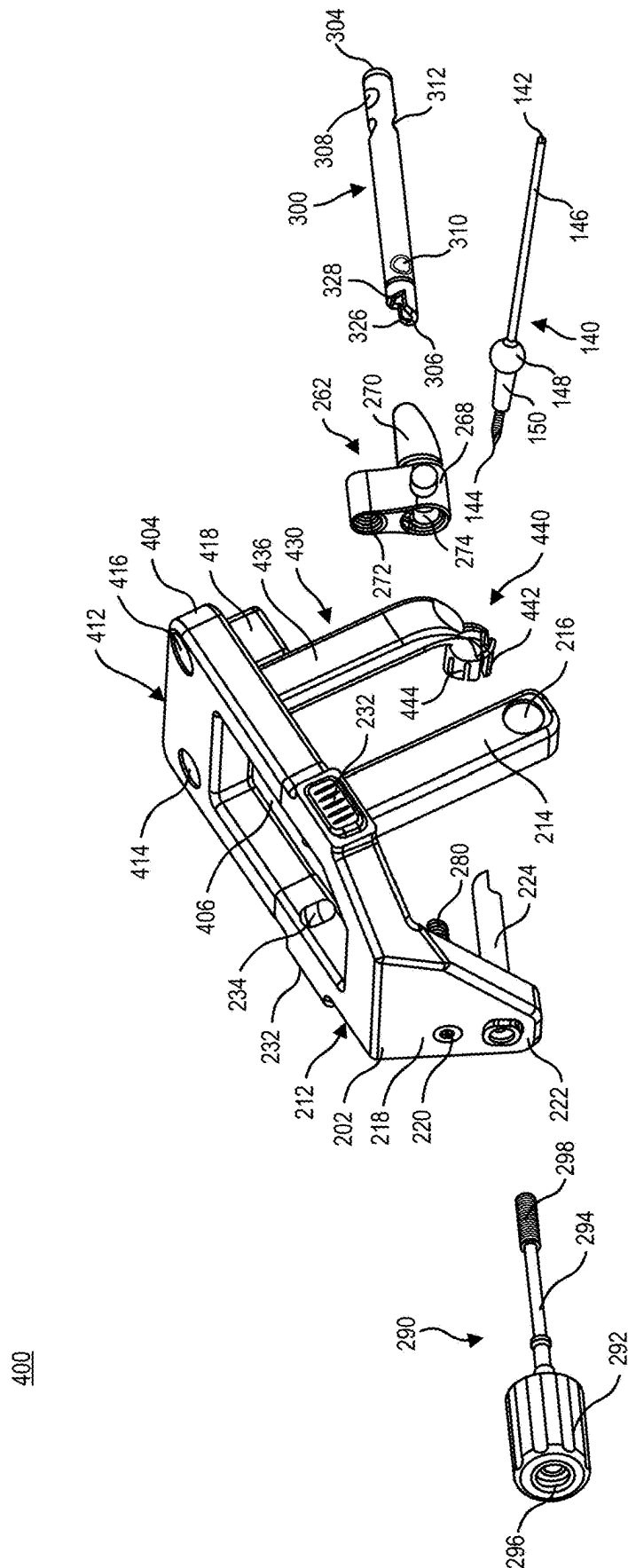


FIG. 25

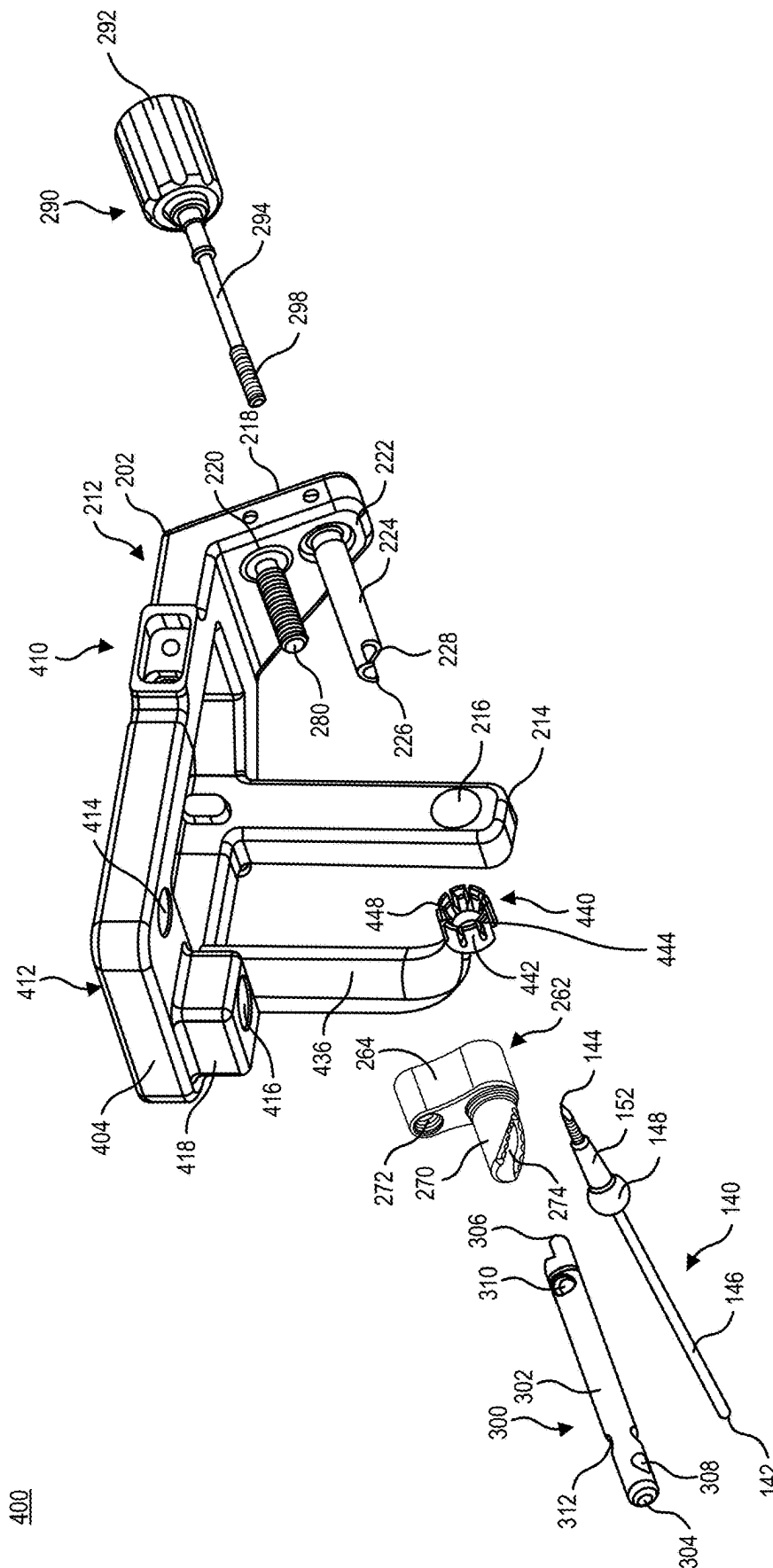


FIG. 26

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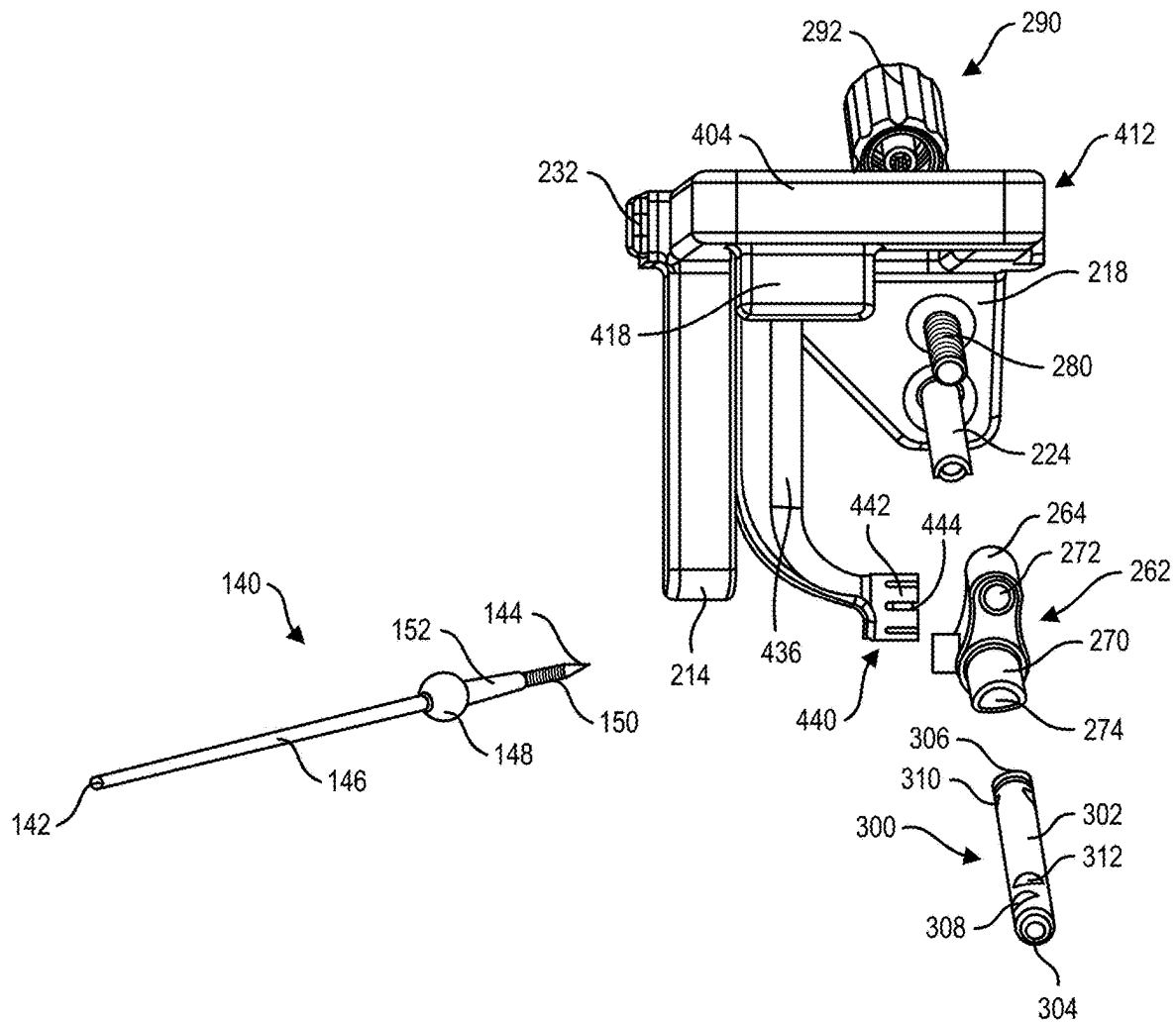


FIG. 27

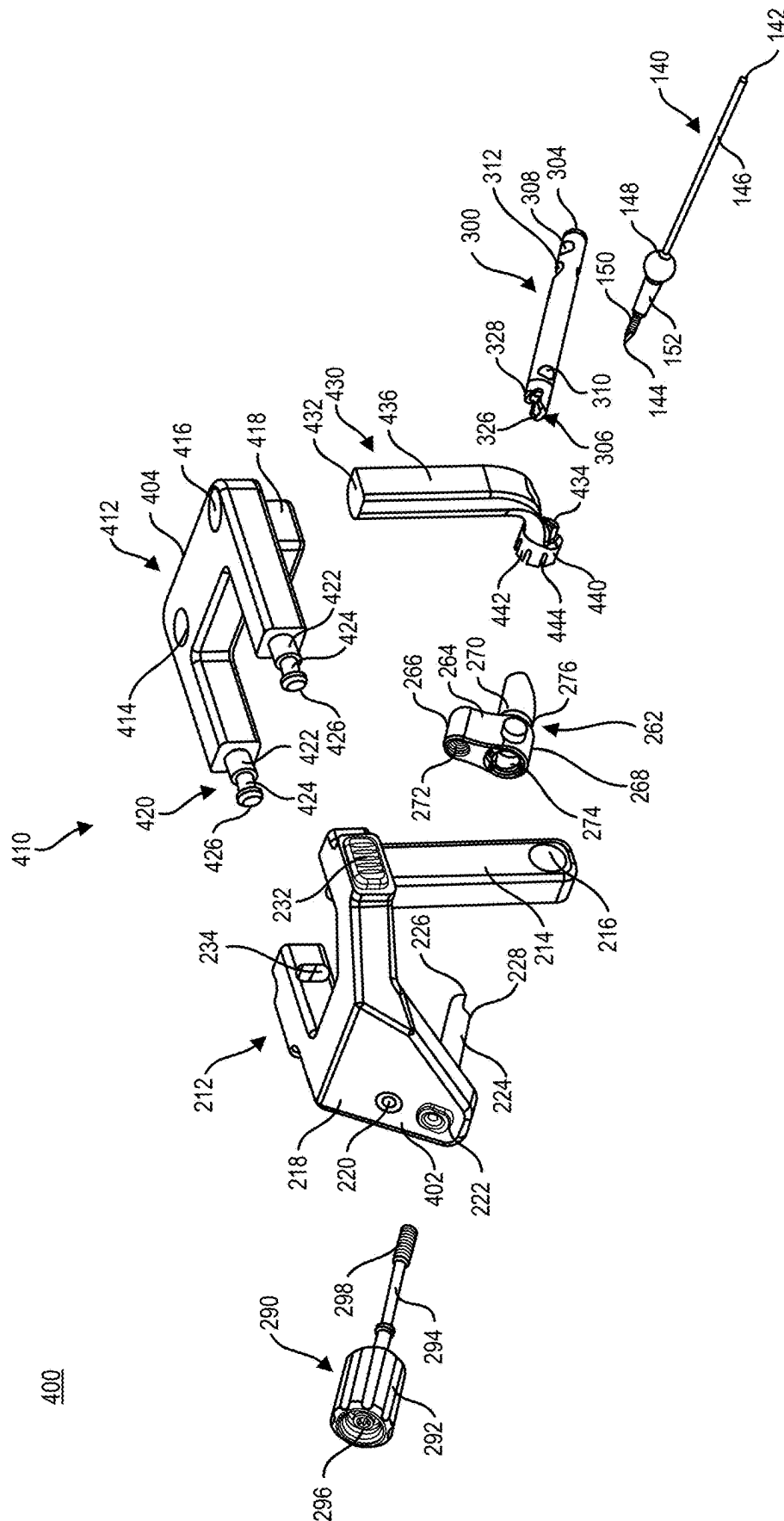


FIG. 28

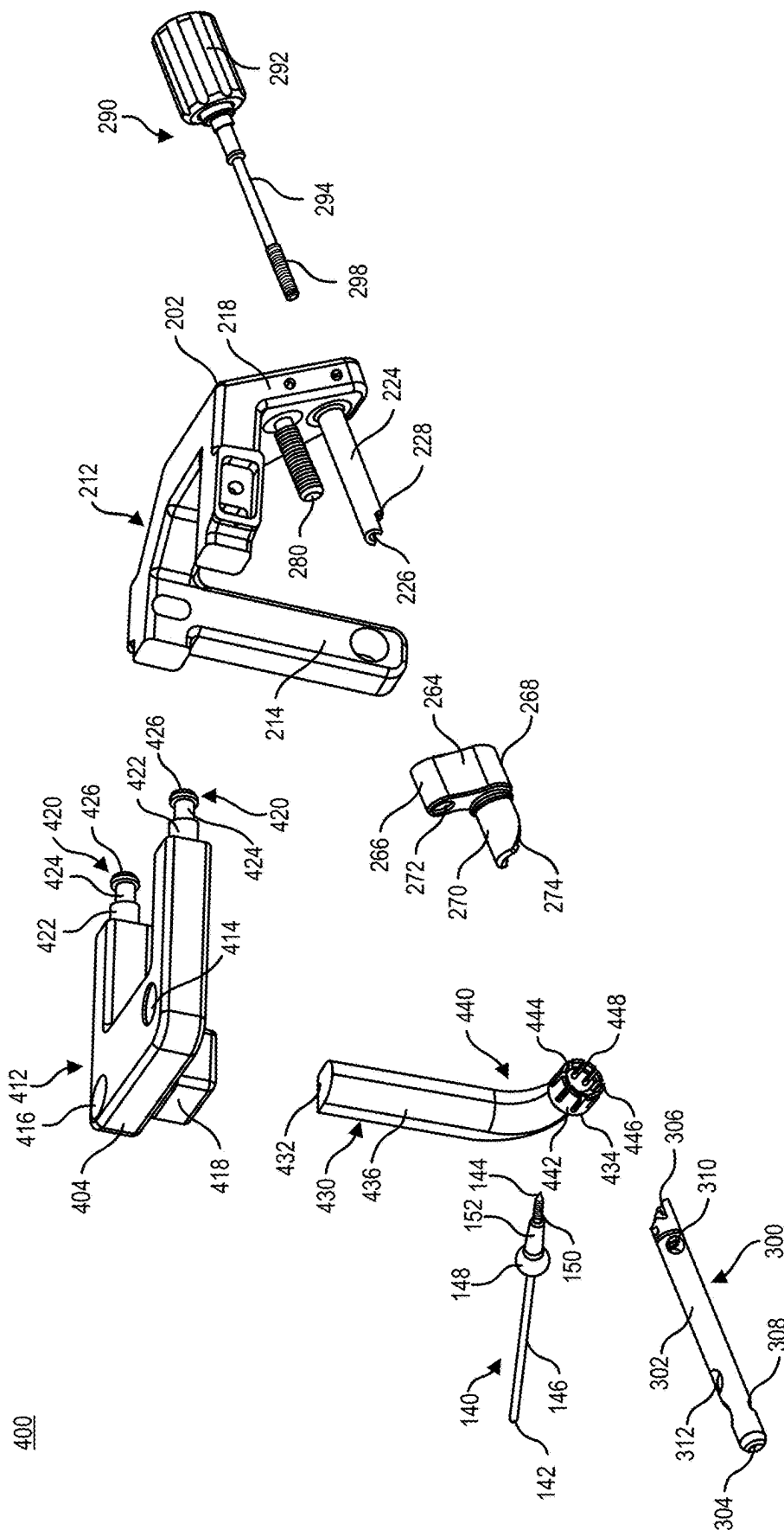


FIG. 29

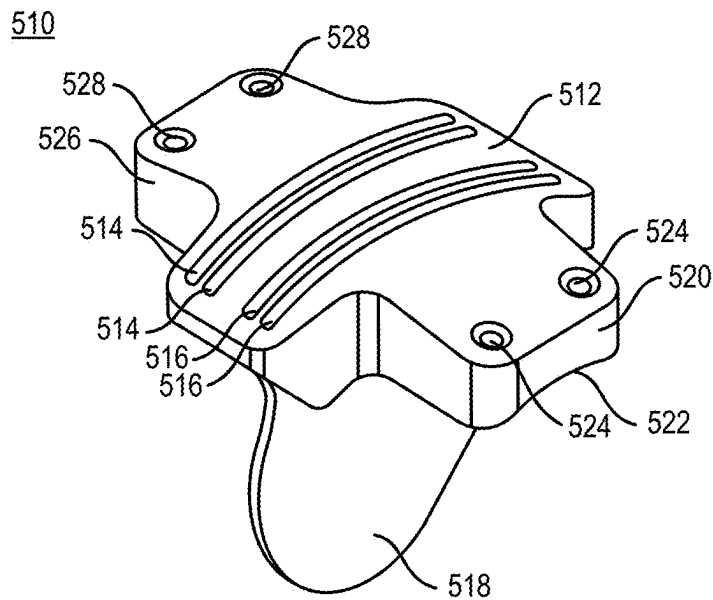


FIG. 30

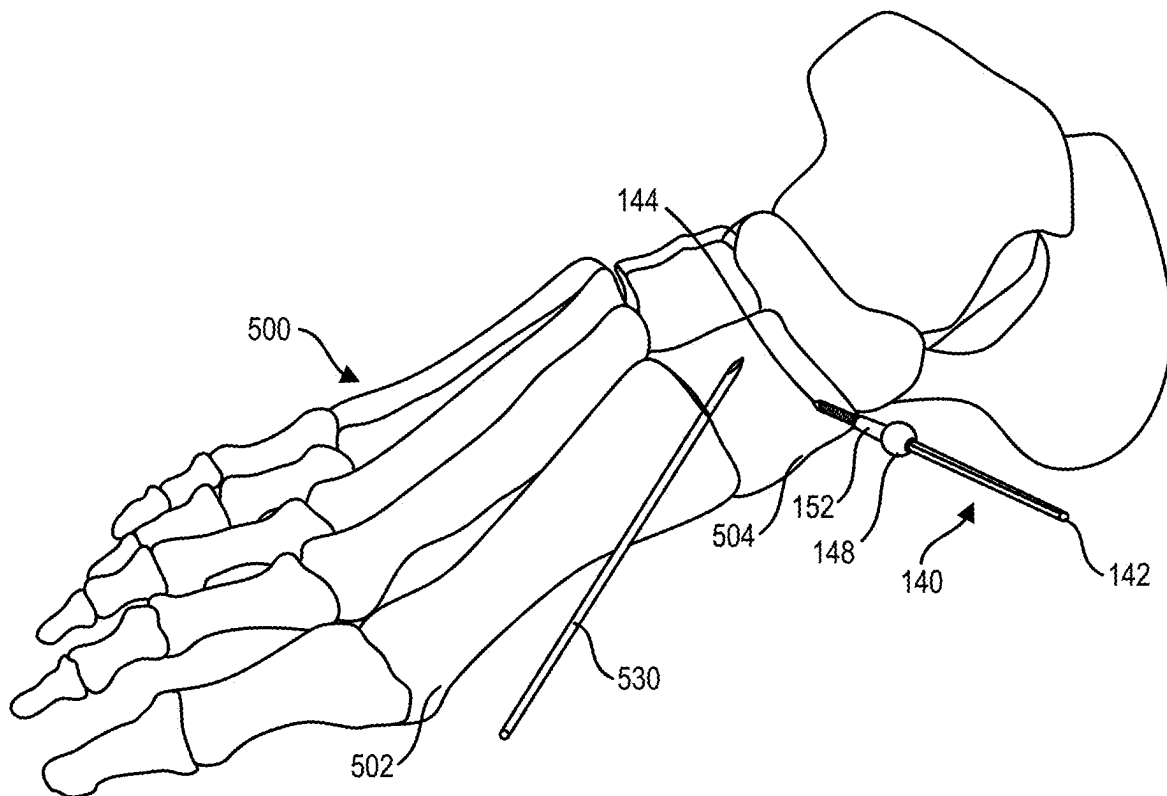


FIG. 31

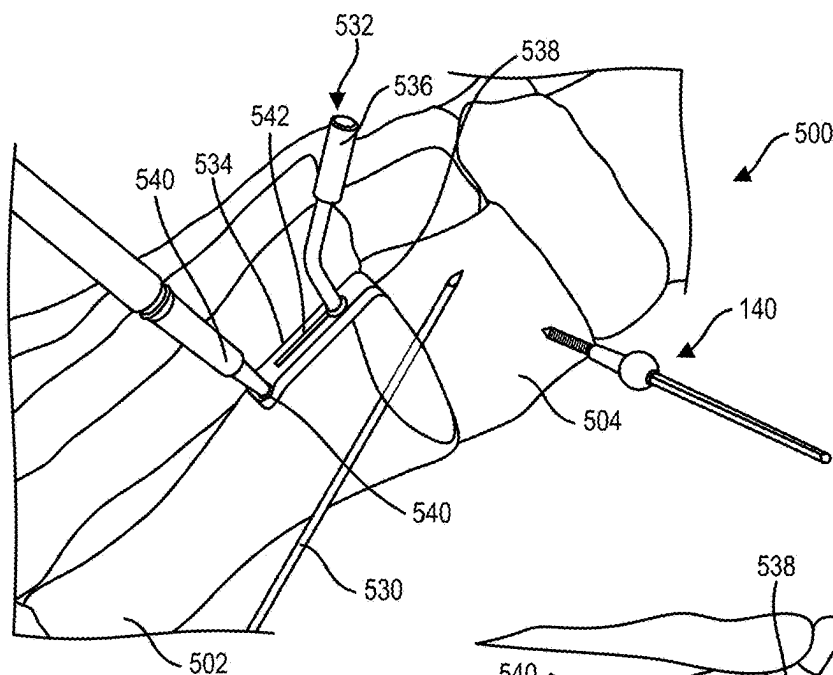


FIG. 32A

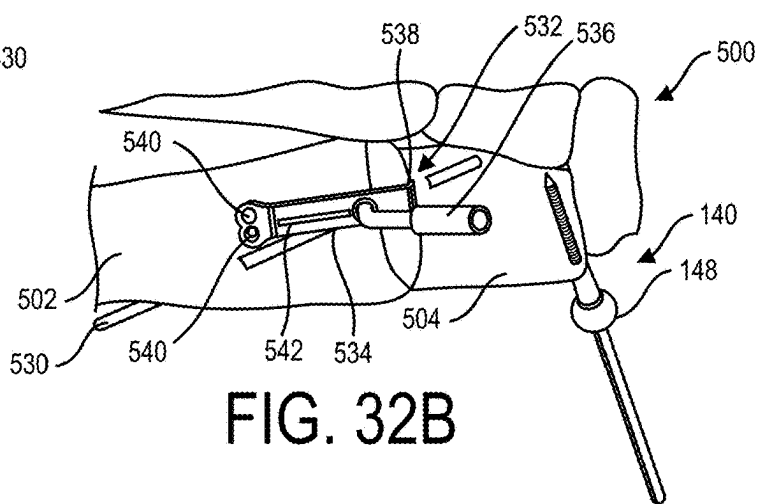


FIG. 32B

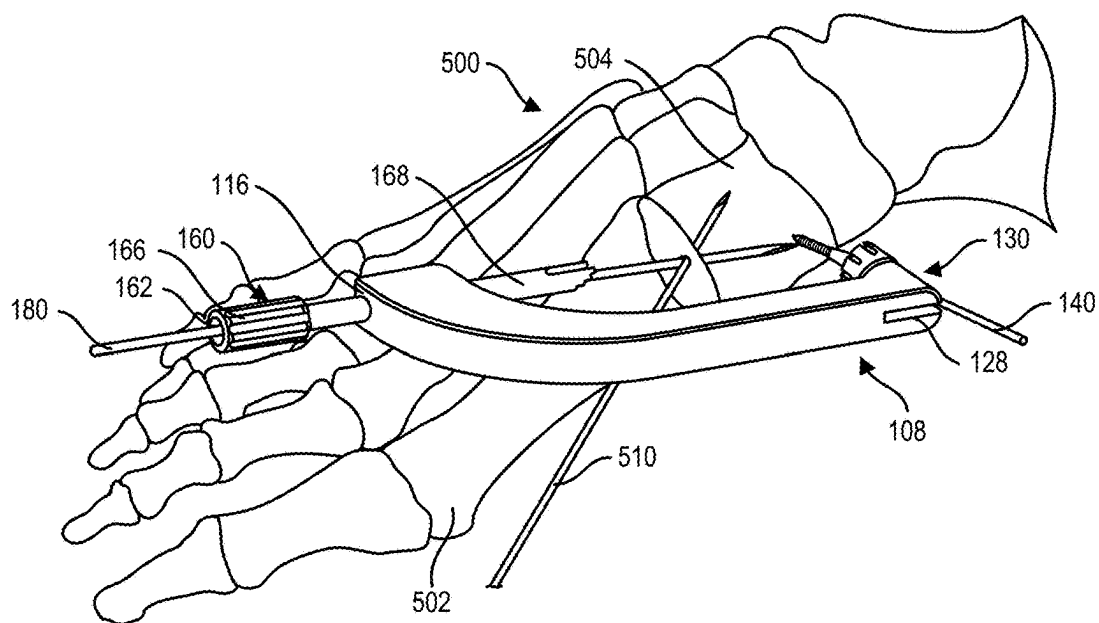


FIG. 33

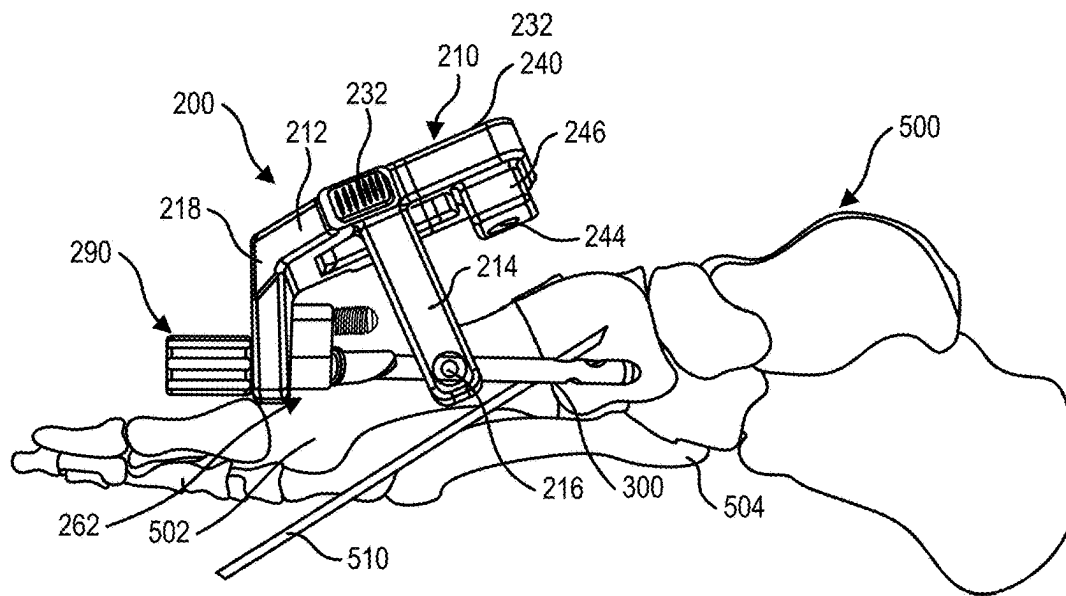


FIG. 34

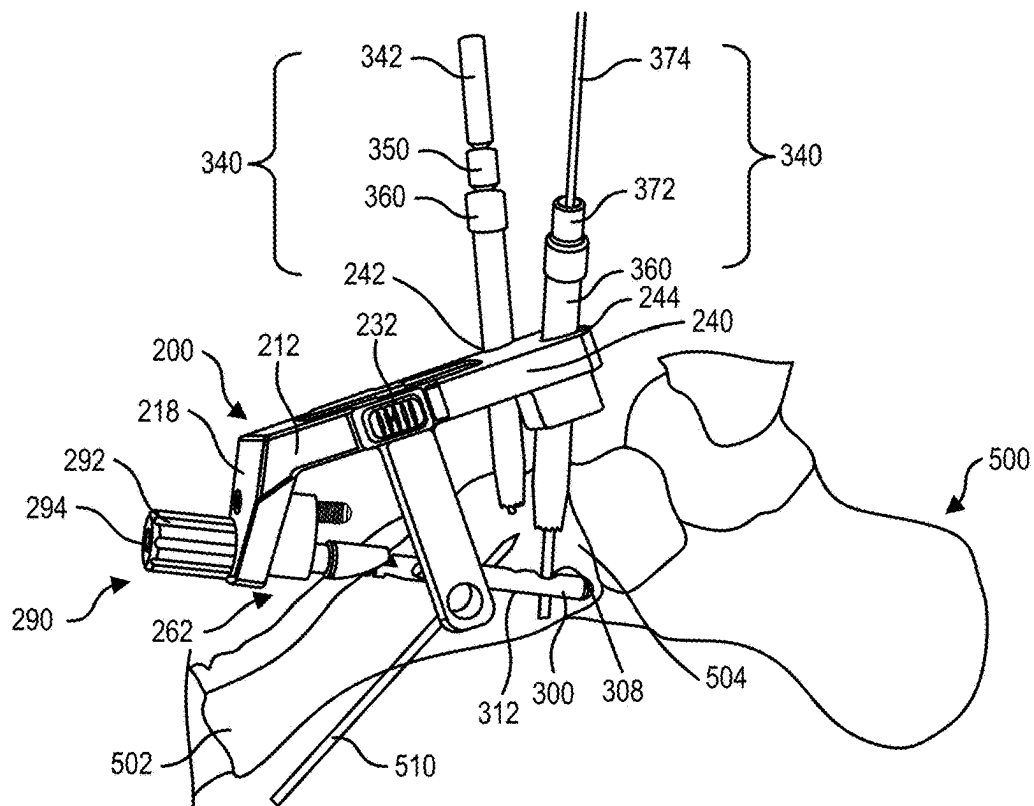


FIG. 35

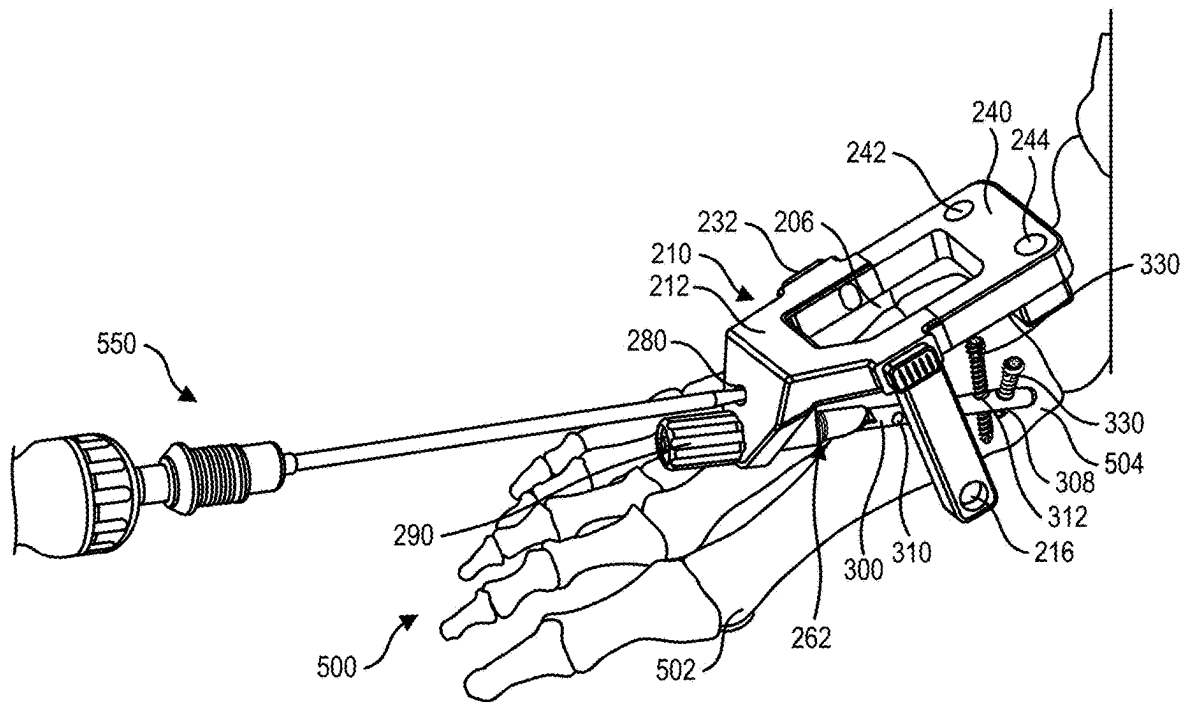


FIG. 36

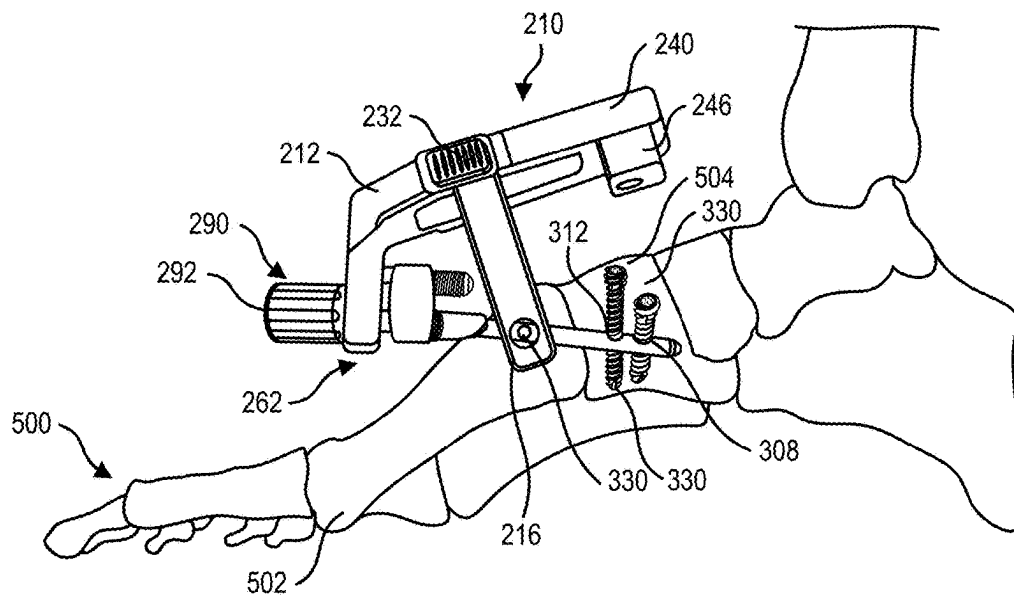


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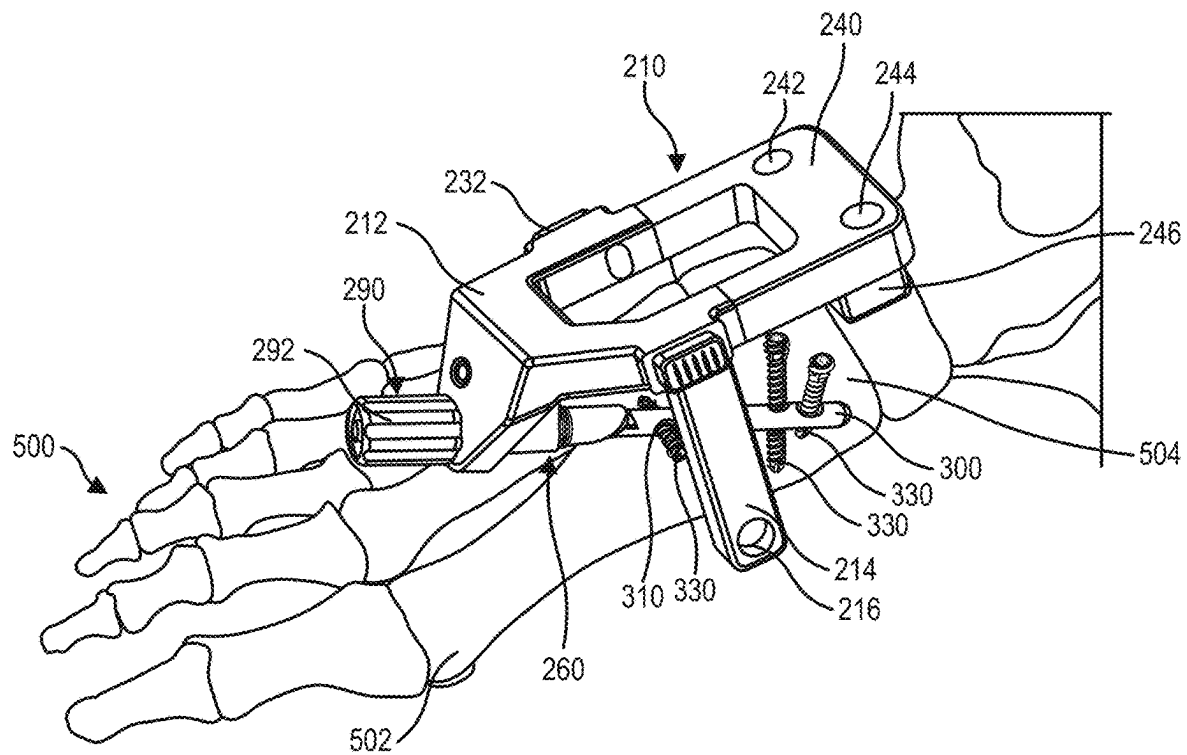


FIG. 38

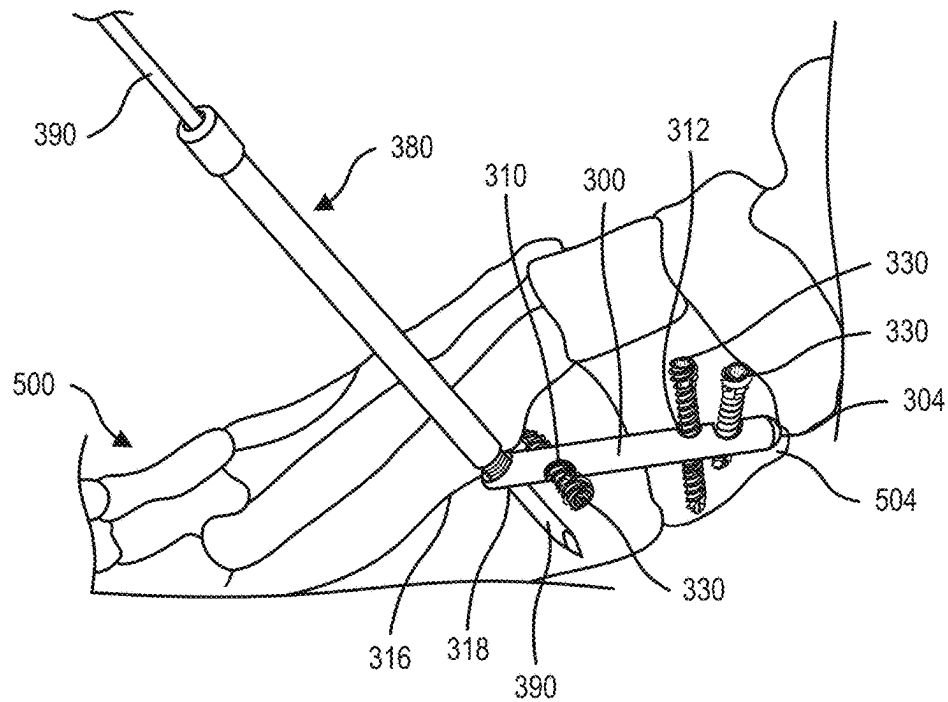


FIG. 39

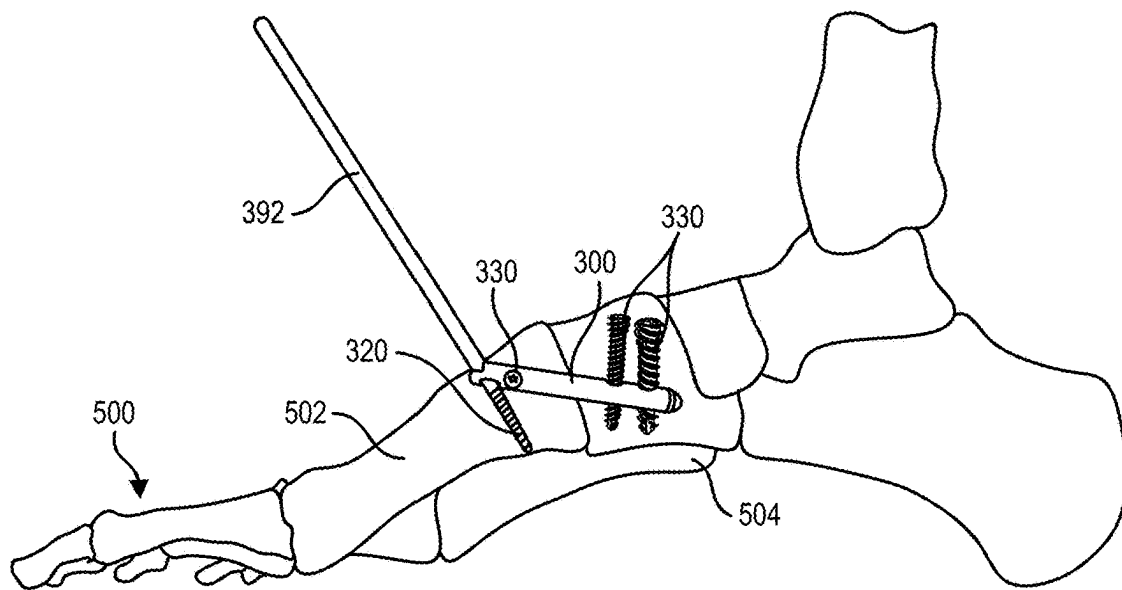


FIG. 40

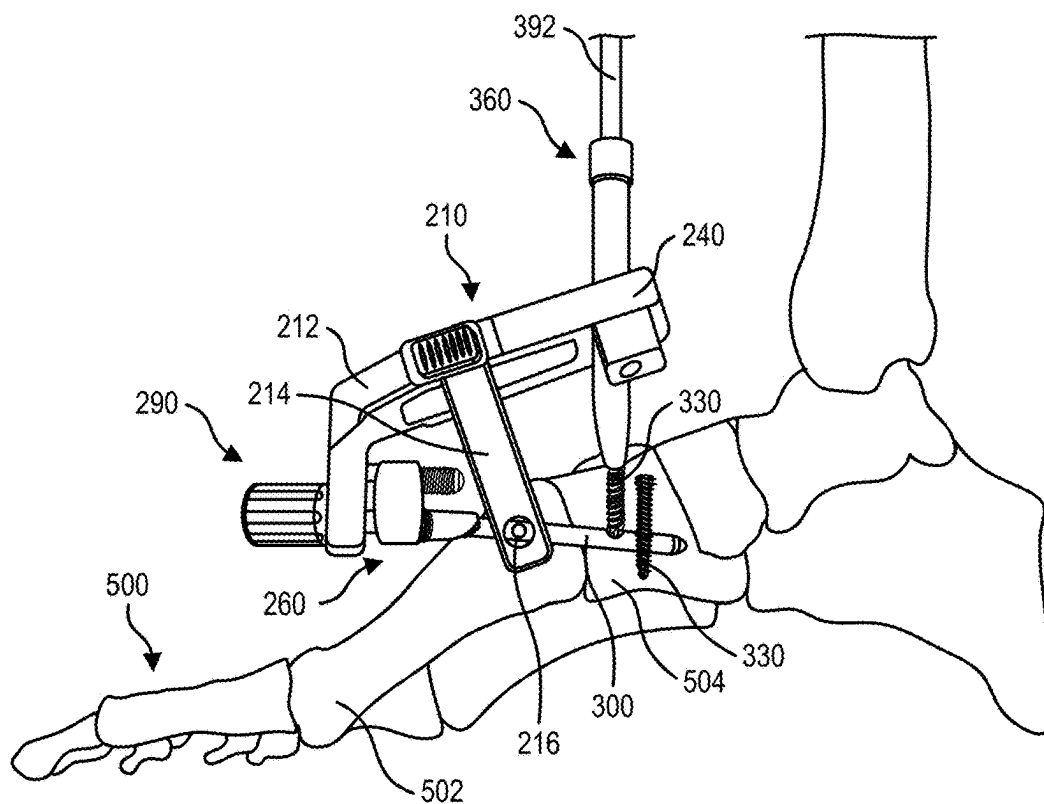


FIG. 41

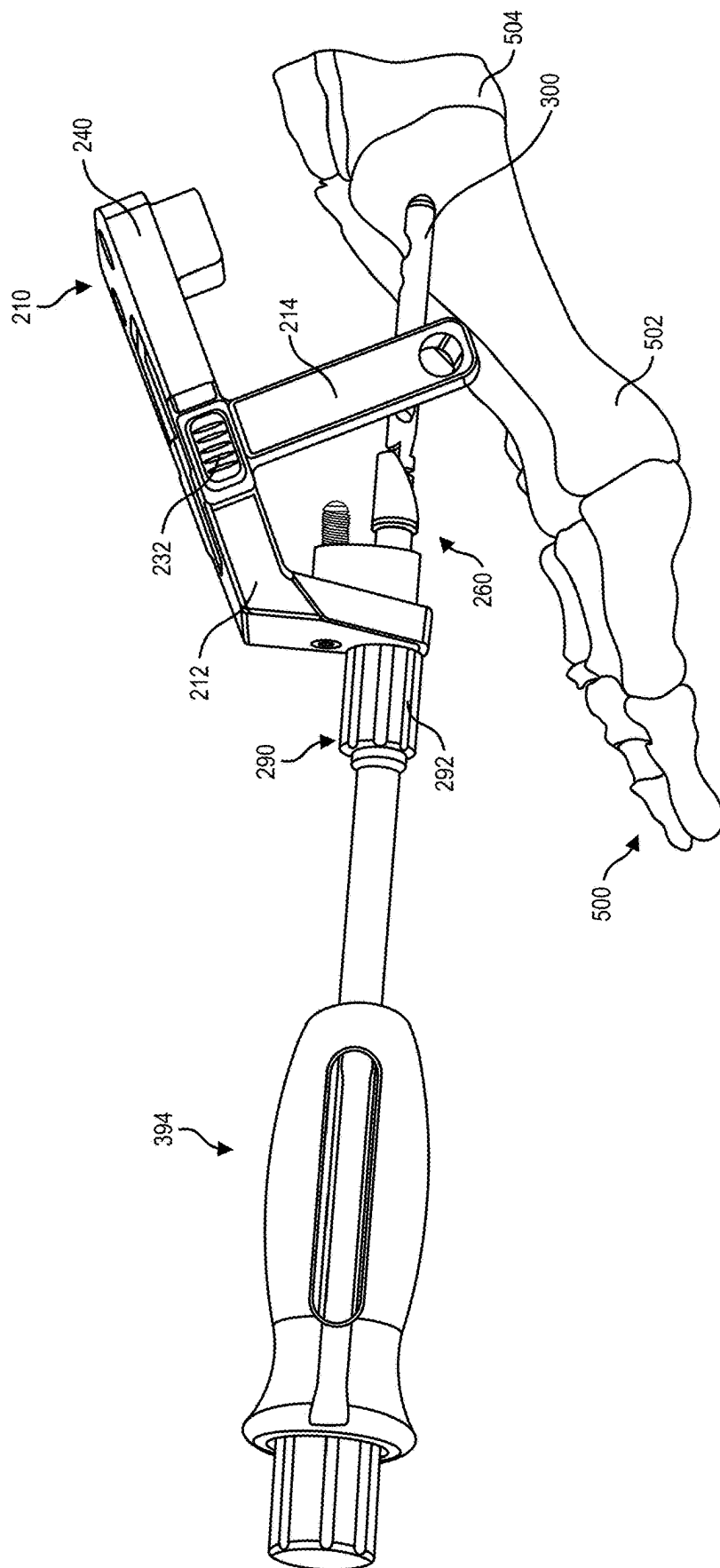


FIG. 42

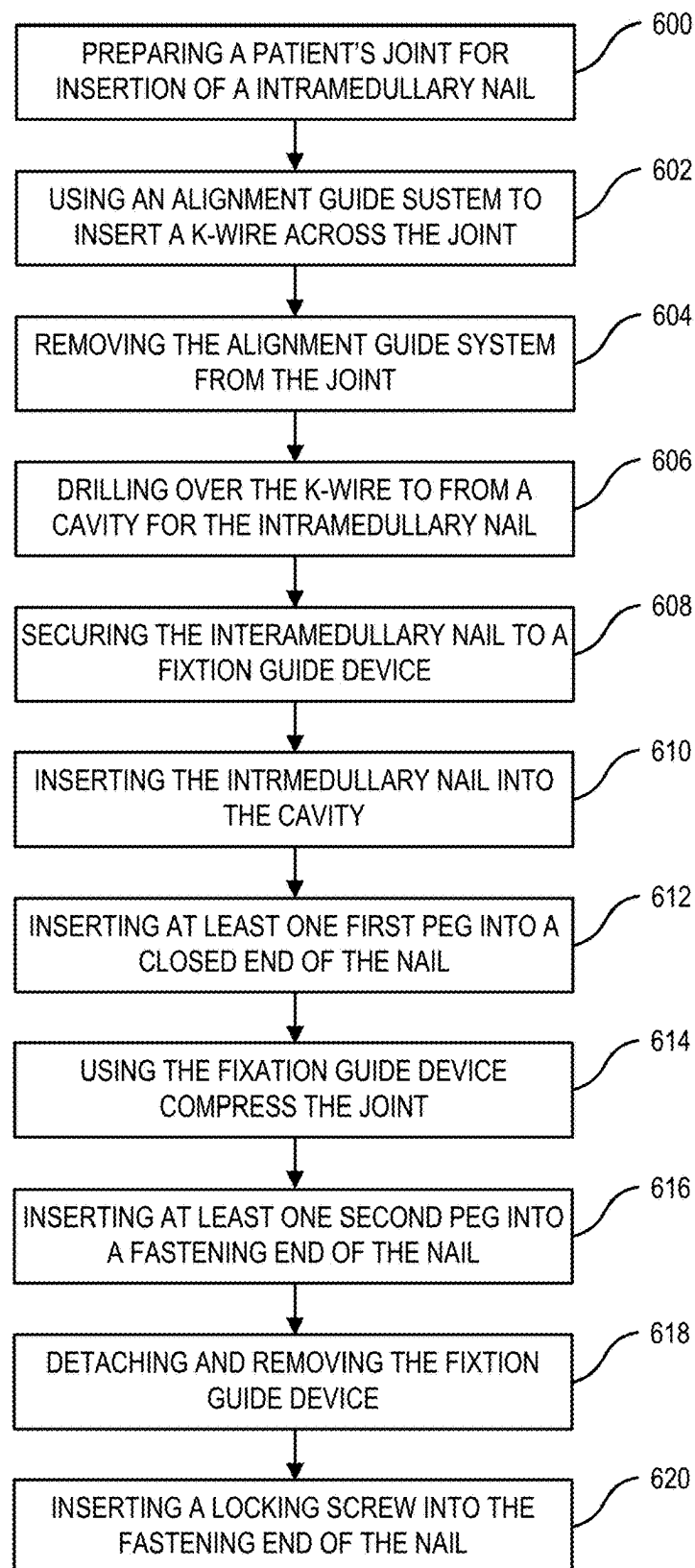


FIG. 43

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INTRAMEDULLARY NAIL ALIGNMENT GUIDES, FIXATION GUIDES, DEVICES, SYSTEMS, AND METHODS OF USE

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. application Ser. No. 17/248,165 filed on Jan. 12, 2021, which issued as U.S. Pat. No. 11,666,345 on Jun. 6, 2023, which is a divisional of U.S. application Ser. No. 15/907,850 filed on Feb. 28, 2018, which issued as U.S. Pat. No. 10,888,338 on Jan. 12, 2021, which is a continuation of PCT Application No. PCT/US2018/020046 filed on Feb. 27, 2018, which claims priority benefit under 35 U.S.C. § 119(e) of U.S. provisional application No. 62/464,175 filed Feb. 27, 2017, which are incorporated herein by reference in their entireties.

TECHNICAL FIELD

The present invention relates generally to general surgery, podiatric, and orthopaedic related to fixation of prepared joint surfaces. More specifically, but not exclusively, the present invention relates to implants, guides, devices, instruments, systems and methods for fixing a joint using an intramedullary nail.

BACKGROUND OF THE INVENTION

The current technology for fusions across two bones utilizes plate and screw fixation that can only be applied on the approach surface. Since the plates are attached outside the fusion zone, the plates will experience bending moments and are limited in their ability to provide uniform compression forces across the fusion site or surfaces. In addition, the plates and screws may create pain due to hardware prominence on the patient's bones. Further, the blood supply to the periosteum may be compromised by attaching a plate to the exterior surface of the bones. Current medial plating systems may experience recurrent hallux valgus and dorsal plating systems may experience plantar gapping over time.

Accordingly, new and improved intramedullary nail alignment guides, fixation guides, devices, and methods which overcome the above-referenced problems and others are needed.

SUMMARY OF THE INVENTION

The present invention is directed toward devices and methods for use in fixation of a patient's joints or for fixation of a fracture. The alignment and fixation guides provide an orientation for insertion of intramedullary nails through a patient's joint or fracture.

In one aspect of the present invention provided herein, is a fixation system. The fixation system including an alignment guide system for forming an opening across a joint and a fixation guide device for inserting an intramedullary nail into the opening.

In another aspect of the present invention provided herein, is an alignment guide system. The alignment guide system including a targeting guide with a first end and a second end, an alignment wire rotatably engaging a second end of the targeting guide, and a guide sleeve insert engaging an opening in the first end of the targeting guide.

In yet another aspect of the present invention provided herein, is a fixation guide device for positioning and inserting an intramedullary nail. The fixation guide device includ-

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ing a frame including a first end and a second end, a compression device slidably coupled to the first end of the frame, and the intramedullary nail secured to a nail attachment apparatus of the frame.

In a further aspect of the present invention provided herein, is a method of inserting an intramedullary nail into two bones for fixation of the two bones. The method may include creating an incision near the two bones and preparing the two bones for fixation. The method may also include securing an alignment guide to at least one of the two bones to position a guidewire across the two bones and removing the alignment guide from the at least one of the two bones. In addition, the method may include drilling over the guidewire to create a cavity for the intramedullary nail that passes through the two bones, wherein the intramedullary nail includes a securing screw hole at a first end of the intramedullary nail and at least two peg holes. The method may further include obtaining a fixation guide device, wherein the fixation guide device includes a frame with at least two drill holes and a compression device. The method may also include attaching the intramedullary nail to the compression device of the fixation guide device and inserting the intramedullary nail into the cavity. Further, the method may include inserting at least one peg into a first bone of the two bones and through at least one first opening in the intramedullary nail and using the at least one first peg and compression device to compress the two bones. The method may also include inserting at least one second peg into a second bone of the two bones and through at least one second opening in the intramedullary nail to secure the two bones in compression and detaching the intramedullary nail from the fixation guide and removing the fixation guide device. In addition, the method may include inserting a locking screw into the securing screw hole. Finally, the method may include closing the incision.

These and other objects, features and advantages of this invention will become apparent from the following detailed description of the various aspects of the invention taken in conjunction with the accompanying drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate embodiments of the invention and together with the detailed description herein, serve to explain the principles of the invention. It is emphasized that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion. The drawings are only for purposes of illustrating preferred embodiments and are not to be construed as limiting the invention.

FIG. 1 is a perspective view from a lateral side of one embodiment of an implant alignment guide system, in accordance with an aspect of the present invention;

FIG. 2 is a top view of the implant alignment guide system of FIG. 1, in accordance with an aspect of the present invention;

FIG. 3 is a top view of a portion of the implant alignment guide system of FIG. 1, in accordance with an aspect of the present invention;

FIG. 4 is an exploded, top perspective view of the implant alignment guide system of FIG. 3, in accordance with an aspect of the present invention;

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FIG. 5 is an exploded, side perspective view of the implant alignment guide system of FIG. 3, in accordance with an aspect of the present invention;

FIG. 6 is an exploded, top view of the implant alignment guide system of FIG. 3, in accordance with an aspect of the present invention;

FIG. 7 is an exploded, bottom view of the implant alignment guide system of FIG. 3, in accordance with an aspect of the present invention;

FIG. 8 is a proximal view of the implant alignment guide system of FIG. 3, in accordance with an aspect of the present invention;

FIG. 9 is a distal view of the implant alignment guide system of FIG. 3, in accordance with an aspect of the present invention;

FIG. 10 is a cross-sectional view of the implant alignment guide of FIG. 1 taken along line 10-10 in FIG. 9, in accordance with an aspect of the present invention;

FIG. 11 is an exploded, side perspective view of a fixation guide system, in accordance with an aspect of the present invention;

FIG. 12 is a side perspective view of the fixation guide of FIG. 11, in accordance with an aspect of the present invention;

FIG. 13 is a medial view of the fixation guide of FIG. 12, in accordance with an aspect of the present invention;

FIG. 14 is a lateral view of the fixation guide of FIG. 12, in accordance with an aspect of the present invention;

FIG. 15 is a top view of the fixation guide of FIG. 12, in accordance with an aspect of the present invention;

FIG. 16 is a bottom view of the fixation guide of FIG. 12, in accordance with an aspect of the present invention;

FIG. 17 is a first end view of the fixation guide of FIG. 12, in accordance with an aspect of the present invention;

FIG. 18 is a second end view of the fixation guide of FIG. 12, in accordance with an aspect of the present invention;

FIG. 19 is an exploded, side view of the fixation guide of FIG. 12, in accordance with an aspect of the present invention;

FIG. 20 shows a perspective view of the fixation guide of FIG. 12, in accordance with an aspect of the present invention;

FIG. 21 is an exploded, first end perspective view of the fixation guide of FIG. 12, in accordance with an aspect of the present invention;

FIG. 22 is an exploded, second end perspective view of the fixation guide of FIG. 12, in accordance with an aspect of the present invention;

FIG. 23 is a partially exploded, lateral view of another embodiment of a fixation guide, in accordance with an aspect of the present invention;

FIG. 24 is a partially exploded, medial view of the fixation guide of FIG. 23, in accordance with an aspect of the present invention;

FIG. 25 is an exploded, lateral view of the fixation guide of FIG. 23, in accordance with an aspect of the present invention;

FIG. 26 is an exploded, medial view of the fixation guide of FIG. 23, in accordance with an aspect of the present invention;

FIG. 27 is an exploded, end view of the fixation guide of FIG. 23, in accordance with an aspect of the present invention;

FIG. 28 is an exploded, perspective first side view of the fixation guide of FIG. 23, in accordance with an aspect of the present invention;

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FIG. 29 is an exploded, perspective second side view of the fixation guide of FIG. 23, in accordance with an aspect of the present invention;

FIG. 30 is a perspective view of a cut guide, in accordance with an aspect of the present invention;

FIG. 31 is a perspective side view of a patient's foot with a k-wire and sphere wire, in accordance with an aspect of the present invention;

FIG. 32A is a perspective view of the embodiment of FIG. 31 with a nail positioning guide positioned on the patient's foot, in accordance with an aspect of the present invention;

FIG. 32B is a top view of the embodiment of FIG. 32A, in accordance with an aspect of the present invention;

FIG. 33 is a perspective side view of the embodiment of FIG. 31 with the alignment guide system of FIG. 1 positioned on the foot and engaging the sphere wire, in accordance with an aspect of the present invention;

FIG. 34 is a perspective side view of the embodiment of FIG. 33 with an intramedullary nail inserted into the patient's foot and coupled to a fixation guide, in accordance with an aspect of the present invention;

FIG. 35 is a perspective side view of the embodiment of FIG. 34 with the fixation guide system coupled to the patient's foot, in accordance with an aspect of the present invention;

FIG. 36 is a perspective top view of the embodiment of FIG. 35 with screw driver, in accordance with an aspect of the present invention;

FIG. 37 is a perspective side view of the embodiment of FIG. 36 showing the threaded pegs being inserted through the intramedullary nail, in accordance with an aspect of the present invention;

FIG. 38 is a perspective top view of the embodiment of FIG. 37 showing all the threaded pegs inserted through the intramedullary nail, in accordance with an aspect of the present invention;

FIG. 39 is a side perspective view of the embodiment of FIG. 38 showing insertion of the locking screw through the intramedullary nail, in accordance with an aspect of the present invention;

FIG. 40 is a side view of the embodiment of FIG. 39 after removal of the fixation guide, in accordance with an aspect of the present invention;

FIG. 41 is a side view of the embodiment of FIG. 40 showing the fixation guide coupled to the intramedullary nail for removal of the nail from the patient's foot, in accordance with an aspect of the present invention;

FIG. 42 is a side view of the embodiment of FIG. 41 showing the intramedullary nail and the fixation guide being removed from the patient's foot, in accordance with an aspect of the present invention; and

FIG. 43 depicts a method of inserting an intramedullary nail into a patient's foot, in accordance with an aspect of the present invention.

DETAILED DESCRIPTION FOR CARRYING OUT THE INVENTION

Generally stated, disclosed herein are implants, guides, devices, instruments, and systems for fixing a joint using an intramedullary nail. Further, methods for implanting and removing an intramedullary nail using the implants, guides, devices, instruments and systems are discussed.

In this detailed description and the following claims, the words proximal, distal, anterior or plantar, posterior or dorsal, medial, lateral, superior and inferior are defined by their standard usage for indicating a particular part or

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portion of a bone or implant according to the relative disposition of the natural bone or directional terms of reference. For example, “proximal” means the portion of a device or implant nearest the torso, while “distal” indicates the portion of the device or implant farthest from the torso. As for directional terms, “anterior” is a direction towards the front side of the body, “posterior” means a direction towards the back side of the body, “medial” means towards the midline of the body, “lateral” is a direction towards the sides or away from the midline of the body, “superior” means a direction above and “inferior” means a direction below another object or structure. Further, specifically in regards to the foot, the term “dorsal” refers to the top of the foot and the term “plantar” refers the bottom of the foot.

Similarly, positions or directions may be used herein with reference to anatomical structures or surfaces. For example, as the current implants, devices, instrumentation and methods are described herein with reference to use with the bones of the foot, the bones of the foot, ankle and lower leg may be used to describe the surfaces, positions, directions or orientations of the implants, devices, instrumentation and methods. Further, the implants, devices, instrumentation and methods, and the aspects, components, features and the like thereof, disclosed herein are described with respect to one side of the body for brevity purposes. However, as the human body is relatively symmetrical or mirrored about a line of symmetry (midline), it is hereby expressly contemplated that the implants, devices, instrumentation and methods, and the aspects, components, features and the like thereof, described and/or illustrated herein may be changed, varied, modified, reconfigured or otherwise altered for use or association with another side of the body for a same or similar purpose without departing from the spirit and scope of the invention. For example, the implants, devices, instrumentation and methods, and the aspects, components, features and the like thereof, described herein with respect to the right foot may be mirrored so that they likewise function with the left foot. Further, the implants, devices, instrumentation and methods, and the aspects, components, features and the like thereof, disclosed herein are described with respect to the foot for brevity purposes, but it should be understood that the implants, devices, instrumentation and methods may be used with other bones of the body having similar structures.

Referring to the drawings, wherein like reference numerals are used to indicate like or analogous components throughout the several views, and with particular reference to FIGS. 1-10, there is illustrated an exemplary embodiment of an alignment guide system 100. The alignment guide system 100 includes an alignment guide or polyaxial targeting guide 108, a pivoting member 140, a guide sleeve insert 160, and a guide wire or k-wire 180. The alignment guide or polyaxial targeting guide 108 includes a body 110 and a pivot assembly 120. The body 110 having a first end 112 and a second end 114. The first end 112 includes a first opening 116 for receiving the guide sleeve insert 160. The first opening 116 may extend entirely through the first end 112 of the body 110 along a longitudinal axis of the alignment guide 108. The second end 114 includes a coupling region 118 and second openings (not shown) extending through the coupling region 118. The coupling region 118 may have a width or thickness smaller than the width or thickness of the rest of the body 110. The second openings may be sized and shaped for receiving fasteners 106 to couple the body 110 to the pivot assembly 120. The body 110 may also have, for example, a curvature or arc as the body extends from the first

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end 112 to the second end 114. The curvature may be positioned, for example, near the distal or first end 112 of the alignment guide 108.

As shown in FIGS. 1-8 and 10, the pivot assembly 120 includes a first end 122 and a second end 124. The first end 122 includes a first slot 126 inset into the first end 122 for receiving the coupling region 118 at the second end 114 of the body 110. The second end 124 includes a pivot slot 128 and a pivoting end 130 for receiving the pivoting member 140. As shown in FIG. 10, the pivot slot 128 may be angled, for example, along the longitudinal axis of the alignment guide 108. The pivot slot 128 may be, for example, narrower on the medial side than the lateral side of the alignment guide 108. The pivoting end 130 may include, for example, a plurality of teeth, protrusions, or extension members 132 alternating with a plurality of grooves or reliefs 134 around the circumference of the pivoting end 130. Each of the plurality of teeth 132 may be, for example, curved on the interior surface of the pivoting end 130 to form a curved region 136. The curved regions 136 on each of the plurality of teeth 132 may form a spherical opening or opening with a circular or round cross-section on the interior surface of the pivoting end 130. The plurality of teeth 132 may also include, for example, a projection or extension 138 positioned near the second end 124 of the pivoting assembly 120 and extending into the spherical opening formed by the curved regions 136 of the teeth 132. The projections 138 provide a retaining surface for coupling the pivoting member 140 to the alignment guide 108, as shown in FIGS. 1-3.

The pivoting member 140 may include a first end 142 and a second end 144, as shown in FIGS. 4-7. The terms “pivoting member,” “sphere wire,” “grip wire,” and “alignment wire” may be used interchangeably herein as the each essentially refer to a wire including a protrusion. The pivoting member 140 may also include a wire portion 146 extending from the first end 142 to a pivot protrusion or spherical member 148. The pivot protrusion 148 may be, for example, spherical or may have a circular or round cross-section and be sized and shaped to match the opening between the plurality of teeth 132 of the pivoting end 130. The pivot protrusion 148 may rotate within the plurality of teeth 132 in the pivoting end 130. The pivoting member 140 may also include an insertion end 150 and a tapered region 152 extending between the pivot protrusion 148 and the insertion end 150. The insertion end 150 may have a pointed tip for insertion into the patient’s foot.

As shown in FIGS. 4-7, the guide sleeve insert 160 may have a first end 162 and a second end 164. The guide sleeve insert 160 may have a handle portion 166 at the first end 162 and a shaft 168 extending from the handle portion 166 to the second end 164. The second end 164 may include a channel 170 extending into the shaft 168. As shown in FIG. 6, the second end 164 may also include a contact portion 172 for contacting a patient’s foot, as shown in FIGS. 2, 3 and 6. The contact portion 172 may include, for example, a plurality of protrusions or other textured surface 174 to assist with contacting the patient’s foot. The guide sleeve insert 160 may be, for example, cannulated to allow for insertion of the guide wire 180. The guide wire 180 may have a first end 182 and a second end 184. The second end 184 of the guide wire 180 may have a pointed or sharp tip for insertion into a patient’s bone.

Referring now to FIGS. 11-22, a fixation guide 200 is shown. The fixation guide 200 includes an outrigger assembly or frame 210 which may be coupled to a compression device 260 and an intramedullary nail 300. The frame 210 may include a first end 202 and a second end 204. The frame

210 may include at least one opening **206** allowing for visualization through the frame **210** with imaging technology, such as x-rays, as seen in FIGS. 15-18. The frame **210** is ideally made of a material that is strong enough to prevent deformation during surgery, for example, a metal, such as aluminum, a composite, such as carbon fiber, or the like. The frame **210** may also be radiolucent to allow for imaging through the frame **210** for determining if correct alignment of the nail **300** was achieved.

In the embodiment depicted in FIGS. 19-22, the outrigger assembly **210** includes a base portion **212** and a guide portion **240**. The base portion **212** includes first or lateral arm **214** and a second or distal arm **218**. The lateral arm **214** extends generally perpendicular to the top surface of the base portion **212** and angled in a proximal-distal direction. The lateral arm **214** includes a first drill hole **216** extending through the arm **214** in a medial-lateral direction. The distal arm **218** extends generally perpendicular to the base portion **212** at the distal end and includes a knob opening or first opening **220**, a nail attachment opening or second opening **222**, and a nail attachment portion or apparatus **224**. The nail attachment portion **224** includes an end with an inverted two step profile including a first step or first nail attachment segment **226** extending a first distance and a second step or second nail attachment segment **228** extending a second distance. The first distance is greater than the second distance.

As shown in FIGS. 20 and 22, the base portion **212** may also include two locking openings **230** for coupling the base portion **212** to the guide portion **240**. The locking openings **230** extend into the base portion **212** from a proximal end. In addition, the base portion **212** may include two buttons or release buttons **232**. The release buttons **232** may include a portion that extends into the locking openings **230** to couple to the guide portion **240**. When depressed the release buttons **232** disengage from the guide portion **240** and allow for removal of the guide portion **240** from the base portion **212**.

The guide portion **240** includes a second drill hole **242** and a third drill hole **244**, as shown in FIGS. 12, 15-18, and 20-22. The second and third drill holes **242**, **244** may each be positioned at different angles relative to the top surface of the guide portion **240**. The third drill hole **244** may extend from the top surface of the guide portion **240** through a projection **246** extending away from a bottom surface of the guide portion **240**. The guide portion **240** may also include two arms **248** for engaging the two locking openings **230** of the base portion **212**. The arms **248** may include a base **250** coupled to the guide portion **240**, a recessed region **252** extending away from the base **250**, and a locking projection **254** extending away from the recessed region **252**. The locking projections **254** having a diameter or size larger than a diameter or size of the recessed region **252**. Multiple guide portions **240** may be provided to engage the base portion **212** and each guide portion **240** may include openings **242**, **244** that correspond to the position of the openings **308**, **312** in different size intramedullary nails **300**. Alternatively, it is also contemplated that the guide portion **240** may include, for example, a plurality of drill holes **242**, **244** as necessary to secure the intramedullary nail **300** across a joint or fracture. The drill holes **242**, **244** may each include, for example, multiple holes spaced a small distance apart or multiple nested or overlapping holes to correspond to the openings **242**, **244** in multiple size intramedullary nails **300**.

When the arms **248** are inserted into the locking openings **230** of the base portion **212**, the arms **248** may extend past the release buttons **232**. The release buttons **232** may include, for example, an extension member **234** extending

through and perpendicular to the locking openings **230**, as shown in FIGS. 21 and 22. The extension members **234** may include, for example, an opening (not shown) that may be aligned with the locking openings **230** in the base portion **212**. For example, when the release buttons **232** are depressed, the openings (not shown) in the extension members **234** may be directly aligned with the locking openings **230** to allow for the arms **248** to be inserted through the openings in the extension members **234**. During insertion, once the locking projections **254** pass through the openings (not shown) in the release buttons **232**, the buttons **232** may be released and the extension members **234** may engage the recessed regions **252** to secure the guide portion **240** to the base portion **212**. During removal of the guide portion **240**, the release buttons **232** may be depressed to align the openings in the extension members **234** with the locking openings **230**. Next, the guide portion **240** may be removed from the base portion **212** by pulling the guide portion **240** until the locking projections **254** pass through the openings (not shown) in the extension members **234** and out of the locking openings **230**.

The compression device **260** slidably mates with the nail attachment portion **224**, as shown in FIGS. 12-14. The compression device **260** includes a compression member **262**, a bolt **280**, and an engagement fastener **290**. As best seen in FIGS. 21 and 22, the compression member **262** has a base **264** with a top end **266** and a bottom end **268**. The compression member **262** also includes a protrusion **270** at the bottom end **268**. A first opening **272** is near the top end **266** for receiving bolt **280**. The first opening **272** is threaded to mate with a threaded end **284** of bolt **280**. A second opening **274** is near the bottom end **268** and passes through the base **264** and protrusion **270**. The second opening **274** is slidably engaged with the nail attachment portion **224**. The proximal end of the compression member **262** is angled from a point between the top and midline of the protrusion **270** to approximately a midpoint **276** of the bottom end **268** of the base **264** of the compression member **262**. The angled portion of the protrusion **270** may include teeth or a texture surface **278** for engaging a patient's bone. The angled portion may have an angle that, for example, mimics or corresponds to the angle of the bone it contacts.

As depicted, after the compression member **262** is slid over nail attachment portion **224**, the compression member **262** may be secured to the base **212** using the bolt **280**, as shown in FIGS. 12-18. The threaded end **284** of bolt **280** is inserted into first opening **220** and the threaded end **284** mates with the threads of the first opening **272**. The bolt **280** may also include a stop member (not shown). The stop member (not shown) prevents bolt **280** from passing through first opening **220**. The bolt **280** may also include a drive opening in the head portion **282** for rotating the bolt **280** to engage and disengage from the compression member **262**. Alternatively, a torque indicating or limiting driver (not shown) including, for example, a torque meter may be used to turn the bolt **280**.

An alternative compression member **262** is also contemplated. The alternative compression member **262** may include a force gauge (not shown) to directly measure the compressive force. The force gauge (not shown) may include gauge indicators or markings (not shown) positioned along the longitudinal axis of the base **264** to show the compressive force being applied.

As shown in FIGS. 21 and 22, the engagement fastener **290** includes a knob **292** coupled to a shaft **294**. The shaft **294** includes a drive opening **296** at a first end and a threaded portion **298** at a second end. Although the threaded portion

298 is shown as only including threads on a portion of the shaft **294**, it is also contemplated that the shaft **294** could be threaded along its entire length. The drive opening **296** may be, for example, hexagonal, square, Phillips or another multi-lobed configuration for coupling with an insertion instrument. The shaft **294** is inserted through the nail attachment opening **222** and the threaded end **298** couples to the intramedullary nail **300** to secure the fixation guide **200** to the nail **300**.

Referring now to FIGS. **11-17** and **19-22**, the intramedullary nail **300** includes a body **302** with a closed end **304**, a fastening end **306**, and openings **308**, **310**, **312**. The openings **308**, **310**, **312** may be, for example, three openings disposed on independent planes and angularly spaced apart relative to each other. By placing the three openings **308**, **310**, **312** at opposing angles oblique to the longitudinal axis of the nail **300**, the amount of longitudinal and rotational movement of the nail **300** is limited. It is also contemplated that the intramedullary nail **300** may include any number of openings **308**, **310**, **312** as may be necessary to secure the nail **300** to a patient's bones. The fastening end **306** includes an insertion opening **316** and an engagement opening **318**. The insertion opening **316** is a threaded opening along a longitudinal axis in the center of the nail **300** parallel to the exterior surface of the nail **300**. The insertion opening **316** is used to secure the nail **300** to the nail attachment portion **224** using an engagement fastener **290**, such as a thumb screw. The engagement opening **318** may include a central axis that is generally transverse the longitudinal axis of the nail **300**. The engagement fastener **290** is inserted through opening **222** and passes through the nail attachment portion **224** before being screwed into the insertion opening **316** to secure nail **300** to the nail attachment portion **224**.

The engagement opening **318** of the nail **300** may be a threaded opening that passes through the exterior surface of nail **300** to secure the nail **300** to the bone using a fastener, such as a locking screw **320**. The locking screw **320** is depicted in FIG. **11** and includes a head **322** with a drive opening (not shown) and a threaded shank **324**. The drive opening (not shown) may be, for example, a Phillips opening, a flat head opening, a hexagonal opening, or other multi-lobed configuration. The threaded shank **324** is designed to engage the patient's bone at the distal end and the threads (not shown) in the engagement opening **318** to secure the nail **300** in place inside the patient's bones.

The nail **300** may also include a fastening end **306** with a shape that corresponds to the shape of the end of the nail attachment portion **224** to create a tight fit between the nail **300** and the attachment portion **224**. As depicted in FIG. **19**, the fastening end **306** may have a two-step profile including a first step or first fastening segment **326** and a second step or second fastening segment **328**. When the nail **300** is secured to the outrigger assembly **210** the first fastening segment **326** aligns with second nail attachment segment **228** and second fastening segment **328** aligns with first nail attachment segment **226**. The engagement fastener **290** may then be inserted to secure the nail **300** to the outrigger assembly **210**. The step profiles of the nail **300** and the nail attachment portion **224** provide additional stability to the nail **300** during insertion into the patient's bone and prevent nail **300** from rotating during insertion of the pegs and screws.

The fixation guide **200** may be used with at least one guide system **340**, as shown in FIG. **11**. The guide system **340** may be used for insertion of pegs or locking screws **330** through the intramedullary nail **300**. The pegs **330** may include a head portion **332** with a drive opening (not shown)

and a threaded shaft **334**. The guide system **340** may include an obturator **342**, drill guide **350**, and a screw guide **360**. The guide system **340** may also include a pin guide **370**, as shown in FIG. **33**. The obturator **342** may include a handle portion **344** coupled to a first end of a shaft portion **346** and a tip **348** at the second end of the shaft portion **346**. The drill guide **350** may include a grip portion **352** at a first end of a shaft portion **354**. The drill guide **350** may also include a cannulation **356** extending from the first end to the second end of the shaft portion **354**. The cannulation **356** may be, for example, sized to receive the obturator **342**. In addition, the drill guide **350** may include teeth **358** at the second end for engagement with the patient's bones. The screw guide **360** may include a grip portion **362** at a first end of the shaft portion **364**. The screw guide **360** may also include a cannulation **366** extending from the first end to the second end of the shaft portion **364**. The cannulation **366** may be, for example, sized to receive the drill guide **350**. In addition, the screw guide **360** may include teeth **368** at the second end for engagement with the patient's bone.

The fixation guide **200** may also include a locking guide **380**, as shown in FIG. **11**. The locking guide **380** may be used for insertion of a locking screw **320** into the intramedullary nail **300**. The locking guide **380** may include a grip portion **382** at a first end of a shaft portion **384**. The locking guide **380** may also include a cannulation **386** extending from the first end to the second end of the shaft portion **384**. The cannulation **386** may be, for example, sized to receive a drill bit or k-wire drill (not shown) for drilling an opening in the bone for insertion of the locking screw **320**. The k-wire drill (not shown) may be used for both drilling the opening in the bone and as a pin for temporary fixation. In addition, the locking guide **380** may include threads **388** at the second end for engagement with the end of the intramedullary nail **300**.

Another embodiment of a fixation guide **400** is shown in FIGS. **23-28**. The fixation guide **400** includes an outrigger assembly or frame **410** which may be coupled to a compression device **260** and an intramedullary nail **300**. The frame **410** may include a first end **402** and a second end **404**. The frame **410** may also include at least one opening **406** allowing for visualization through the frame **410** with imaging technology, such as x-rays, as seen in FIGS. **23-25**. The frame **410** is ideally made of a material that is strong enough to prevent deformation during surgery, such as a metal, while also being radiolucent to allow for imaging through the frame **410** to determine if correct alignment of the nail **300** was achieved. The frame **410** may be made of, for example, carbon fiber.

As shown in FIGS. **28-29**, the frame **410** includes a base portion **212** and a guide portion **412**. The base portion **212** may be as described above with reference to FIGS. **11-22**, which will not be described again here for brevity sake. The guide portion **412** includes a second drill hole **414** and a third drill hole **416**, as shown in FIGS. **23-26**, **28** and **29**. The second and third drill holes **414**, **416** may each be positioned at different angles relative to the top surface of the guide portion **240**. The third drill hole **244** may extend from the top surface of the guide portion **412** through a projection **418** extending away from a bottom surface of the guide portion **412**. Multiple guide portions **412** may be provided to engage the base portion **212** and each guide portion **412** may include openings **414**, **416** that correspond to the position of the openings **308**, **312** in different size intramedullary nails **300**. Alternatively, it is also contemplated that the guide portion **412** may include, for example, a plurality of drill holes **414**, **416** as necessary to secure the intramedullary nail **300** across

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a joint or fracture. The drill holes **414**, **416** may each include, for example, multiple holes spaced a small distance apart or multiple nested or overlapping holes to correspond to the openings **414**, **416** in multiple size intramedullary nails **300**.

The guide portion **412** may also include an alignment arm **430**, as shown in FIGS. **23-29**. The alignment arm **430** has a first end **432** and a second end **434**. The first end **432** may be coupled or secured to a bottom surface of the guide portion **412**. The second end **434** may include a pivoting end **440** for receiving the pivoting member **140**. The pivoting end **440** may include, for example, a plurality of teeth, protrusions, or extension members **442** alternating with a plurality of grooves or reliefs **444** around the circumference of the pivoting end **440**. Each of the plurality of teeth **442** may be, for example, curved on the interior surface of the pivoting end **440** to form a curved region **446**, as shown in FIGS. **26** and **29**. The curved region **446** on each of the plurality of teeth **442** may form a spherical opening or opening with a circular or round cross-section on the interior surface of the pivoting end **440**. The shape formed by the curved regions **446** may, for example, match the shape of the pivot protrusion **148** of the pivoting member **140**. The plurality of teeth **442** may also include, for example, a projection or extension **448** positioned near the plantar end of the alignment arm **430** and extending into the spherical opening formed by the curved regions **446** of the teeth **442**. The projections **448** provide a retaining surface for coupling the pivoting member **140** to the alignment arm **430**, as shown in FIGS. **23** and **24**. The pivoting member **140** is of the type described above with reference to FIGS. **11-22**, which will not be described again here for brevity sake.

As shown in FIGS. **28** and **29**, the guide portion **412** may include two arms **420** for engaging the two locking openings **230** of the base portion **212**. The arms **420** may include a base **422** coupled to the guide portion **412**, a recessed region **424** extending away from the base portion **422**, and a locking projection **426** extending away from the recessed region **424**. The locking projections **426** having a diameter or size larger than a diameter or size of the recessed region **424**. The arms **420** are configured to engage the locking openings **230** in the base portion **212**. When the arms **420** are inserted into the locking openings **230** of the base portion **212**, the arms **420** may extend past the release buttons **232**. The release buttons **232** may be of the type described above with reference to FIGS. **23-26** and **28** and will not be described again here for brevity sake. In addition, when the release buttons **232** are depressed, the openings (not shown) in the extension members **234** may be directly aligned with the locking openings **230** to allow for the arms **420** to be inserted through the openings in the extension members **234**. During insertion, once the locking projections **426** pass through the openings (not shown) in the release buttons **232**, the buttons **232** may be released and the extension members **234** may engage the recessed regions **424** to secure the guide portion **412** to the base portion **212**. During removal of the guide portion **412**, the release buttons **232** may be depressed to align the openings in the extension members **234** with the locking openings **230**. Next, the guide portion **412** may be removed from the base portion **212** by pulling the guide portion **412** until the locking projections **426** pass through the openings (not shown) in the extension members **234** and out of the locking openings **230**.

The compression device **260** may be of the type described above with reference to FIGS. **12-14**, which will not be described again here for brevity sake. In addition, the engagement fastener **290**, as shown in FIGS. **25-29**, may be of the type described above with reference to FIGS. **11** and

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19-21. Thus, the fixation guide **400** may be assembled in the same way as described above, which will not be described again here for brevity sake.

The surgical method for inserting an intramedullary nail **300** into a patient's foot **500** across a joint is shown in FIGS. **30-43**. As shown in FIG. **43**, the method may include preparing a patient's joint for insertion of an intramedullary nail **600**. The method may also include using an alignment guide system to insert a k-wire across the patient's joint **602** and removing the alignment guide system from the joint **604**. The method may further include drilling over the k-wire to form a cavity for the intramedullary nail **606** and securing the intramedullary nail to a fixation guide device **608**. Next, the method may include inserting the intramedullary nail into the cavity **610** and inserting at least one first peg into a closed end of the intramedullary nail **612**. In addition, the method may include using the fixation guide device to compression the joint **614** and inserting at least one second peg into a fixation end of the intramedullary nail **616**. Further, the method may include detaching and removing the fixation guide device **618** and inserting a fastening screw into the fastening end of the intramedullary nail **620**.

Referring now to FIGS. **30-42**, the method may be described in greater detail. For example, the method may include making an incision over a joint, for example, a medial or dorsomedial incision over a first tarsometatarsal joint. Then, tissue dissection may be performed to expose the joint. Optionally, the cartilage may be resected and removed from the joint, then subchondral bone preparation may be performed. The cartilage resection may be performed, for example, using a cut guide **510**, as shown in FIG. **30**. The cut guide **510** may include a body or base member **512**. The body **512** may include at least one first slot **514** on a first side of the body **512** and at least one second slot **516** on a second side of the body **512**. The cut guide **510** may also include a fin or paddle **518** extending from a bottom surface of the body **512**. The fin **518** is sized and shaped to fit into a joint between two bones, for example, the metatarsal and cuneiform bones. The bottom surface of the body **512** may be, for example, anatomically curved to mate with the surface of the dorsal surface of the patient's bones. The guide **510** may also include a first arm **520** extending from a first end of the body **512** and a second arm **526** extending from a second end of the body **512**. The first arm **520** may extend in a proximal direction while the second arm **526** may extend in a distal direction. The first arm **520** may have a bottom surface **522**, which may be anatomically curved to correspond to the patient's bones. The first arm **520** may also include at least one hole **524** for securing the cut guide **510** to a first bone for cutting a portion of the first bone. The second arm **526** may include at least one hole **528** for securing the cut guide **510** to a second bone for cutting a portion of the second bone. If a bone graft or implant (not shown) is needed to restore length to the patient's toe after the resection, the bone graft or implant (not shown) may then be inserted into the joint. The bone graft or implant (not shown) may be, for example, shaped to match the shape of the adjacent bones **502**, **504**.

The method may also include inserting a k-wire **530**, for example, through the metatarsal bone **502** and into the medial cuneiform bone **504** across the first tarsal-metatarsal joint, as shown in FIG. **31**. The k-wire **530** may be placed from plantar medial distal to dorsal lateral proximal. Next, a pivoting member or sphere wire **140** may be obtained and placed at the proximal plantar medial aspect of the medial cuneiform **504**, as shown in FIG. **31**. Then, the pivoting end **130** of the alignment guide **108** may be coupled to the

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pivoting protrusion **148** of the pivoting member **140**. Once coupled the alignment guide **108** may pivot about the pivoting protrusion **148** of the pivoting member **140**.

Referring now to FIGS. **32A** and **32B**, an optional nail positioning guide **532** is shown. The nail positioning guide **532** may be used to assist with placement of a k-wire **180**. The nail positioning guide **532** may include a base **534** and a handle **536** extending away from a top surface of the base **534** near the first end. The first end of the base **534** also includes a projection or leg **538** extending from a bottom surface of the base **534**. The base **534** may also include at least one alignment opening **540** near the first end and an alignment marking **542** extending at least partially between the first end and the second end. The method may optionally include inserting the projection **538** into the central lateral aspect of the first tarsal-metatarsal joint. Once the projection **538** is inserted into the joint, the at least one alignment opening **540** is positioned on the patient's bone **502** at the desired start position for the k-wire **180**. Next, a pilot hole may be formed in the patient's bone **502** through the at least one alignment opening **540** for appropriately lining up the k-wire for insertion.

Next, if not already coupled to the pivoting member **140**, the alignment guide **108** may be coupled to the pivoting member **140**, as shown in FIG. **33**. A guide sleeve insert **160** may then be inserted into the first opening **116** in the distal end of the guide **108** and placed into contact with the patient's bone **502**. The trajectory of the guide sleeve insert **160** is then aligned for the desired starting position of the k-wire **180**, which matches the desired trajectory of the nail **300**. If the nail positioning guide **532** was used, the distal end of the guide sleeve insert **160** may be aligned to the pivot hole in the bone **502**. Specifically, the channel **170** of the guide sleeve insert **160** may be aligned over the pivot hole in the bone **502**. Then, the k-wire **180** may be inserted through the guide sleeve insert **160** and into the patient's bone **502** until it contacts the pivoting protrusion **148** of the pivoting member **140** or reaches the proximal plantar aspect of the medial cuneiform **504**. Placement of the k-wire **180** may then be confirmed using fluoroscopy. Once the desired placement of the k-wire **180** is achieved, the sleeve insert **160** may be removed from the k-wire **180** and guide **108** removed from the pivoting member **140**. After the sleeve insert **160** and guide **108** are removed, the length of the k-wire **180** may be measured with a nail depth gauge (not shown).

A cannulated drill bit (not shown) of a drill may then be inserted over the k-wire **180** and drilled into the bones **502**, **504** until the pivoting member **140** is contacted. Once an opening is drilled over the k-wire **180**, the pivoting member **140** may be removed from the patient's bone **504**. Next, a fixation guide **200** may be obtained and assembled including selecting and attaching a guide portion **240** that corresponds to the measured size of the inserted nail **300**. The nail **300** may also be coupled to the fixation guide **200**. The engagement fastener **290** is then inserted into the guide **210** and coupled to the selected nail **300**. The nail **300** may be coupled to the guide **210** by turning the engagement fastener **290** in a clockwise direction to thread the engagement fastener **290** into the inside of the nail **300**. Then the assembled alignment guide **108** may be inserted into the drilled hole, as shown in FIG. **34**. The nail **300** may be inserted until the contoured piece on the frame **210** is flush with the dorsal aspect of the first metatarsal. Fluoroscopy may then be used to confirm the size of the nail **300** and the placement of the nail. The pivoting member **140** may then be removed.

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Referring now to FIGS. **11** and **35**, an obturator **342** may be inserted into the drill guide **350** and the drill guide **350** may be inserted into a screw guide **360**. The assembled guide system **340** may then be inserted into the second drill hole **242**, as shown in FIG. **35**. A pin guide system **370** may then be inserted into the third drill hole **244**. The pin guide system **370** may include a screw guide **360**, a k-wire guide **372** sized to fit inside of the screw guide **360**, and a k-wire **374** sized to fit within the k-wire guide **372**. The k-wire **374** may be inserted through the k-wire guide **372**, into the patient's bone **504** and through an opening **308** in the nail **300**, as shown in FIG. **35**. Next, the obturator **342** may be removed from the drill guide **350** and a drill bit (not shown) or k-wire drill may be inserted through the drill guide **350** to drill into the patient's bone **504**. Then, the drill guide **350** may be removed from the screw guide **360** and a depth gauge (not shown) is inserted through the screw guide **360** to measure for selecting a first threaded peg **330**. The selected threaded peg **330** may then be inserted through the screw guide **360** into the bone **504** through the opening **312** in the nail **300**, as shown in FIG. **36**. After the threaded peg **330** is inserted the screw guide **360** may be removed from the fixation guide **200**.

Next, the k-wire **374** and k-wire guide **372** may be removed from the screw guide **360** in the third drill hole **244**. A drill guide (not shown), such as guide **350**, may then be inserted into the screw guide **360**. A drill bit (not shown) may then be inserted through the drill guide (not shown) to drill into the patient's bone **504**. The drill guide (not shown) may then be removed from the screw guide **360** and depth gauge (not shown) may be inserted through the screw guide **360** to measure for selecting a second threaded peg **330**. The selected threaded peg **330** may be inserted through the screw guide **360** into the bone **504** through the opening **308** in the nail **300**, as shown in FIG. **36**. Once the threaded pegs **330** are inserted through the nail openings **308**, **312**, the screw guides **360** may be removed from the second and third drill holes **242**, **244**. Then, the temporary fixator or k-wire **510** may be removed from across the joint.

With continued reference to FIG. **36**, a solid driver **550** may be used to tighten the bolt **280** of the fixation guide **200** to translate the compression member **262** creating compression across the arthrodesis site. Compression is created between the medial cuneiform bone **504** secured to the nail **300** with pegs **330** and the compression member **262** contacting and translating the metatarsal bone **502** into closer contact with the medial cuneiform bone **504**, as shown in FIG. **37**. The compression member **262** pushes the metatarsal bone **502** into contact with the medial cuneiform bone **504** along the longitudinal axis of the nail **300**. The compression device **260** provides controlled compression across the fusion joint to produce the proper surface contact stress between the metatarsal bone **502** and medial cuneiform bone **504** to promote osteosynthesis and enable proper healing. The protrusion **270** of the compression device **260** includes a convex surface for applying a central axis force and contact compression to bones having different topographical morphologies of bone. The end of the protrusion **270** is angled for oblique contact with the patient's bone and the teeth or textured surface **278** provide a gripping surface to assist with gripping the bone during compression of the joint.

Once the desired compression is achieved across the joint, a third peg **330** may be inserted through the nail opening **310**, as shown in FIG. **37**. The third peg **330** may be inserted by, for example, inserting a coupled obturator **342**, drill guide **350** and screw guide **360** through the first drill hole

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216 of the fixation guide 200. The obturator 342 may then be removed from the drill guide 350 and a drill bit (not shown) may be inserted through the drill guide 350 to drill into the patient's bone 502. Then, the drill guide 350 may be removed from the screw guide 360 and a depth gauge (not shown) may be inserted through the screw guide 360 to measure for selecting the third threaded peg 330. The selected threaded peg 330 may then be inserted through the screw guide 360 into the bone 502 through the opening 310 in the nail 300, as shown in FIGS. 37 and 38. After the threaded peg 330 is inserted the screw guide 360 may be removed from the fixation guide 200.

Next, as shown in FIG. 39, the fixation guide 200 may be removed from the patient's foot 500. The fixation guide 200 may be removed by turning the knob 292 of the engagement fastener 290. As the knob 292 is turned counterclockwise, the threaded portion 298 of the engagement fastener 290 disengages the insertion opening 316 in the nail 300. Then, the fixation guide 200 may be removed from the patient's foot 500, as shown in FIG. 39. Next, a locking guide 380 may be placed into the insertion opening 316 of the nail 300, as shown in FIG. 39. A drill bit or k-wire drill 390 may then be used to drill an opening through the metatarsal 502. A depth gauge (not shown) may be inserted through the locking guide 380 to measure the size locking screw 320 needed, as shown in FIG. 39. Then, the depth gauge (not shown) and locking guide 380 may be removed and the locking screw 320 may be inserted through the nail opening 318 using a driver 392, as shown in FIG. 40. The locking screw 320 is inserted into the metatarsal 502 and the placement and size are then confirmed using fluoroscopy.

The method may further include a removal or revision procedure. The removal procedure may include removing the locking screw 320 from the intramedullary nail 300 using the driver 392, as shown in FIG. 40. Next, a fixation guide 200 may be selected include the guide portion 240 that corresponds to the nail 300 being removed. The frame 210 of the fixation guide 200 may be coupled to the nail 300 and the guide portion 240 may be inserted and locked to the frame 210, as shown in FIG. 41. An engagement fastener 290 may then be inserted into the frame 210 and turned to thread into the insertion opening 316 in the nail 300. Once the fixation guide 200 is secured to the nail 300, a screw guide 360 may be inserted into the second drill hole 242, as shown in FIG. 41. A driver 392 may then be inserted into the cannulation 366 of the screw guide 360 and inserted into the head of the first peg 330. The driver 392 may be rotated to remove the first peg 330 from the nail opening 312 and out through the screw guide 360. Next, the screw guide 360 may be removed from the second drill hole 242 and inserted into the third drill hole 244. The driver 392 may then be inserted through the screw guide 360 and inserted into the head of the second peg 330. Next, the second peg 330 may be removed from the nail opening 308 and out through the screw guide 360. The screw guide 360 may then be removed from the third drill hole 244 and inserted into the first drill hole 216. The driver 392 may then be inserted through the screw guide 360 and inserted into the head of the third peg 330. Next, the third peg 330 may be removed from the nail opening 310 and out through the screw guide 360. Finally, a slaphammer 394 may be attached to the engagement fastener 290 of the fixation guide 200. Then the slaphammer 394 may be used to back the nail 300 out of the patient's foot 500.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms

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as well, unless the context clearly indicates otherwise. It will be further understood that the terms "comprise" (and any form of comprise, such as "comprises" and "comprising"), "have" (and any form of have, such as "has", and "having"), "include" (and any form of include, such as "includes" and "including"), and "contain" (and any form of contain, such as "contains" and "containing") are open-ended linking verbs. As a result, a method or device that "comprises," "has," "includes," or "contains" one or more steps or elements possesses those one or more steps or elements, but is not limited to possessing only those one or more steps or elements. Likewise, a step of a method or an element of a device that "comprises," "has," "includes," or "contains" one or more features possesses those one or more features, but is not limited to possessing only those one or more features. Furthermore, a device or structure that is configured in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

The invention has been described with reference to the preferred embodiments. It will be understood that the architectural and operational embodiments described herein are exemplary of a plurality of possible arrangements to provide the same general features, characteristics, and general system operation. Modifications and alterations will occur to others upon a reading and understanding of the preceding detailed description. It is intended that the invention be construed as including all such modifications and alterations.

What is claimed is:

1. An intramedullary nail, comprising:
a body, comprising:
a closed end at a first end; and
a fastening end at a second end, the fastening end comprising:
a first fastening segment with a first length and positioned on a bottom of the body; and
a second fastening segment with a second length and positioned on a top of the body, wherein the first length is longer than the second length, and wherein the second fastening segment has an angled engagement surface relative to a surface of the fastening end and extending into the body from the second end toward the first end; and
at least two openings extending into a side of the body, wherein the at least two openings are disposed on independent planes and angularly spaced apart from each other.
2. The intramedullary nail of claim 1, wherein the fastening end comprises:
an insertion opening extending into the body along a longitudinal axis of the intramedullary nail from the second end; and
an engagement opening extending into the body of the intramedullary nail transverse to the longitudinal axis of the intramedullary nail.
3. The intramedullary nail of claim 2, wherein the insertion opening extends through the first fastening segment and the second fastening segment.
4. The intramedullary nail of claim 3, wherein the engagement opening extends through the first fastening segment.
5. The intramedullary nail of claim 4, wherein the engagement opening extends through the body adjacent to the second fastening segment.
6. The intramedullary nail of claim 1, wherein the first fastening segment has a first end at the fastening end and a second end where the first fastening segment engages the second fastening segment, wherein the angled engagement

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surface of the second fastening segment has a first end at the fastening end and a second end where the second fastening segment engages the first fastening segment, and wherein the second end of the first fastening segment is positioned closer to the closed end of the intramedullary nail than the first end of the second fastening segment.

7. A compression system, comprising:
an intramedullary nail, comprising:

a body, comprising:

a closed end at a first end; and

a fastening end at a second end, the fastening end comprising:

a first fastening segment with a first length and positioned on a bottom of the body; and

a second fastening segment with a second length and positioned on a top of the body, wherein the first length is longer than the second length, and wherein the second fastening segment has an angled engagement surface relative to a surface of the fastening end and extending into the body from the second end toward the first end; and

at least two openings extending into a side of the body, wherein the at least two openings are disposed on independent planes and angularly spaced apart from each other; and

a compression device for engaging the intramedullary nail.

8. The compression system of claim 7, wherein the fastening end comprises:

an insertion opening extending into the body along a longitudinal axis of the intramedullary nail from the second end; and

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an engagement opening extending into the body of the intramedullary nail transverse to the longitudinal axis of the intramedullary nail.

9. The compression system of claim 8, wherein the compression device comprises:

a compression member configured to engage the fastening end of the intramedullary nail, wherein the compression member comprises:

a base with a top end and a bottom end; and

a protrusion extending away from a first side at the bottom end of the base, wherein a bottom surface of the protrusion is angled from a point between a top and a midline of the protrusion to a position near a midpoint of a bottom end of the protrusion along a longitudinal axis of the protrusion.

10. The compression system of claim 9, wherein the compression member further comprises:

a first opening positioned near the top end and proximal to the protrusion, wherein the first opening receives a bolt; and

a second opening positioned near the bottom end and extending through the protrusion.

11. The compression system of claim 10, wherein the compression member further comprises:

an engagement fastener extending through the compression member and engaging the insertion opening of the intramedullary nail, wherein the engagement fastener comprises:

a knob at a first end;

a shaft extending away from the knob to a second end; and

a threaded end for coupling to the intramedullary nail.

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